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PROGRAMMING DEMYSTIFIED

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PROGRAMMING DEMYSTIFIED

SEMINAR GUIDE

2ND EDITION

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Stress free adaptation

- "It is impossible not to see that the stress concept adequately describes pathological states only…" (Yerokhin)

Adaptive reactions' characteristics (Based on Garkavi, Ukolova, Kvakina, Verkhoshanskaya)					
	Adaptive reactions				
	Training	**Calm activation**	**High activation**	**Over-activation**	**Stress**
Load	Low	Medium	Optimal	Critical	Excessive
Performance	Increases from left to right and becomes highly unstable in the stress reaction				
When desirable	Reentry	Training		Peaking, competition	Never
Plastic processes					
Intensity	Low	High	Very high		
Dominance	Anabolic			Balanced	Catabolic
Energetics					
Intensity	Low	High; aerobic metabolism is hyperactive		Very high	Extremely high; aerobic metabolism is depressed and glycolysis is upregulated
Dominance	Accumulation	Balanced		Mild depletion	Depletion
Psycho-emotional state					
Mood	Calm, inert	Calm, active	Very active, eager, optimistic	Very active, aggressive	Depressed, anxious, [aggressive]
Sleep	Adequate	Good	Excellent	Disturbed	Poor
Appetite				Good	

"There is a belief that an athlete must reach high levels of fatigue to reach an anabolic effect. In principle, this is correct, as fatigue is often associated with destruction of cellular structures, which strongly intensifies synthetic processes—even though they have a compensatory nature.[1] But destruction is not

[1] "Compensatory"—as opposed to "developmental"—protein synthesis repairs the damage and uses up plastic, energetic, and endocrine resources.

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the only stimulus of synthetic processes. Another…is creating in muscle cells such conditions that accelerate adaptive synthetic processes…and that absolutely does not have to be accompanied by pronounced damage to muscles' morphological structures."

"Hence in choosing training means and methods it is very important to evaluate catabolic and anabolic potential of each exercise. Thus, one of the principles of training session construction is the "principle of factoring in the anabolic/catabolic potential of an exercise." (Myakinchenko & Selouyanov)

- "That which does not kill us makes us not stronger, but less sensitive." (Verkhoshanskaya) In response to a strong but non-lethal stimulus the CNS develops defensive inhibition, its sensitivity decreases.

Continuity

- Repetition of the same stimuli is the main condition for triggering adaptive reactions. The organism perceives varied stimuli as "static" and does not respond to them with a system wide reaction.

- "It is one of the most important training principles: do not change the rhythm of the loads… The biological system must get exact information about the direction of adaptation." (Arbeit)

  "…dynamic stereotypes are developed when certain stimuli are repeated in a strict order and with strict intervals between them." (Pavlov)

  Thus, a distinct rhythm of loads within microcycles and mesocycles is called for when one is using progressive overload[2], e.g.:

  - Lift at the same time of the day, doing the same exercises on particular days
  - Train every other day or follow another rhythmical schedule
  - Lay out a clear progression pattern within a mesocycle
  - Take time to adapt to a particular training system

Accommodation

- "The timing of heavy resistance protocols in micro- and mesocycles is dominated by two main ideas. One is to allow adequate recovery between exercise periods, and the other is to find a proper balance between the steadiness of the training stimulus (to call forth an adaptation) and its variability (to avoid premature accommodation and staleness)." (Zatsiorsky)

  On one hand, training must be same: the principles of continuity and specificity.

  On the other, it must be different due to the *law of accommodation*.

  Accommodation is overcome with changes, qualitative or quantitative.

[2] But not variable overload such as Plan Strong™ and Built Strong.

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1) Qualitative

Exercises and training methods.

An established practice is changing one and/or the other every mesocycle—although there are many successful precedents when the exercises and the methods remain constant and only the quantitative parameters are varied.

The most reliable approach to changing exercises without losing the transfer is adhering to specialized variety.

2) Quantitative

Volume, intensity, and various temporary parameters.

Possible changes:

- Increasing the load
- Changing the loading pattern (set and rep scheme, etc.)
- Decreasing the load to resensitize the organism to weaker stimuli and to replenish used up adaptative reserves

A successful tactic is making quantitative changes every training session while making qualitative changes every mesocycle.

Beware that the more changes you make, the harder it is to analyze the effectiveness of particular training tactics. A successful program must have many constants and a limited number of variables.

Reactivity and resistance to a load

There are two types of relationships between the athlete's organism and the training load: *reactivity* and *resistance*.

Resistance (stability, immunity) refers to the ability to handle a given load and is synonymous with the medical term.

Reactivity (sensitivity, responsiveness) refers to the potential of the given load to cause a desired adaptive reaction.

The two characteristics ride a teeter-totter: when one goes up, the other comes down. When you introduce a greater training stimulus, the organism is highly reactive or responsive to it. But the resistance is low: it is easy to overtrain. As you stay with the given load for some time, you are getting progressively more resistant and less reactive.

"Periodization of sport training manifests itself in a periodic restoration of the initial ratio of the system's sensitivity and stability to the load on each successive, higher level…" (Mishoustin)

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Strength training load parameters

1. **Specificity**
2. **Novelty**
3. **Volume**
4. **Intensity**
5. **Reps and effort**
6. **Rest** (frequency; duration and nature of rest between reps, sets, and series; session duration)
7. **Exercise dynamics**
8. **Preceding load**
9. **Following load**
10. **Load dynamics**

- Interaction of the load parameters (intensity, volume, etc.) is so complex that it is impossible to isolate the relative role of any one of them.

- Individual limit load tolerance is largely dependent on the endocrine system reserves: the sympatho-adrenal system and hypothalamic-pituitary-adrenocortical systems.

- Understand the difference between the *external load* and the *internal load*.

  On one hand, purely physical-mechanical measurements like weight, speed, distance, etc. ("external"). On the other, internal changes in the body in response to this external work: higher heart rate, etc. Internal load describes the organism's reaction to the external load. "[Both internal and external load criteria] are important for evaluating and directing loads… External load criteria must be factored in because they provide guidance for setting achievable and necessary load values. The organism's responses are then compared to them and loads are rationed accordingly." (Matveev)

1. Specificity

- The more similar the exercise to the competitive event, the more stressful it is.

- Weightlifting or powerlifting exercises classification:

  I. Competition lifts proper
  II. Special preparatory exercises *(specialized variety)*
  III. Additional development exercises

  Share of category I to category II:

  - Beginner—100%
  - Intermediate—70-80%
  - Advanced—up to 50%

  Change the ratio of category I to II every mesocycle, e.g., 70:30, 80:20, 75:25. Reduce the number and share of assistance exercises in the competition month, especially the last two weeks. Change the category II and III exercises every mesocycle.

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- *Specialized variety exercises* mimic the lift the athlete wants to improve. Specialized variety exercises build strength by providing a new stimulus while still keeping it specific to the target lift and by recruiting new motor units. *Conjugate training method.* (Dyachkov)

  The motor program of the competition lift remains, the changes take place in one or more of the surface features. A short list of possible changes:

  1) Stance
  2) Head position
  3) Grip
  4) Incline
  5) Dynamics (speed, pauses)
  6) Added bands, chains, etc.
  7) Implement
  8) Range of motion
  9) Displacement of the weight
  10) Assistance gear

2. Novelty

- The organism is more reactive and less resistant to a novel stimulus.

3. Volume

- Usually measured with NBL (the number of barbell lifts); "NL" if another implement is used.

- Primary function of volume: producing stable long term adaptations (both morphological and neurological).

  "Up to a certain volume of training (called the saturation threshold) there exists a direct dependence of increase [in results] on the volume." (Viru)

  For experienced weightlifters there is a high correlation between the NBL @ ≥70% 1RM and the competitive results.

- Secondary functions of volume: strengthening the connective tissues; increasing CP and glycogen capacity.

- The following volume guidelines for experienced strength athletes are based on Soviet weightlifting experience but are applicable to many training systems:

Daily NL per class of exercise, ≥70% 1RM

- **10-20**—minimal
- **20-30**—optimal
- **30-50**—maximal

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4. Intensity

- In strength training, "intensity" describes how heavy the resistance is, compared to one's maximum. It may be measured in % 1RM or RM.

- In addition to the *repetition maximum*, some coaches track the *series maximum* (SM) that indicates the heaviest weight one can lift for a given set and rep scheme. E.g., 225x5/5 SM.

- Primary function of intensity: raise the performance "ceiling."

- Secondary function: peaking.

- When 70, 80, and 90% 1RM intensities are compared, 90% weights build strength most rapidly. However, these gains are unsustainable; 80% intensity develops more strength over a longer term.

- The intensity varies depending on the chosen training system and the specific point in the training cycle.

- The following considerations for >90% 1RM and heavier lifts are applicable to all systems.

    - The stress experienced in the 91-100% 1RM intensity zone is "psychophysical"—as opposed to the "physical" stress of training in the lower zones.

    - Exceeding the individual optimal number of >90% 1RM lifts delays progress at best and leads to overtraining at worst, while an insufficient number of such lifts retards strength gains.

    - Younger, less experienced lifters better tolerate a higher volume of lifts in the top zone: low level athletes can handle twice as much as the elite. Lighter lifters can tolerate twice as heavier athletes. There are no differences between athletes taking and not taking anabolic steroids.

    - The optimal number of lifts in the top intensity zone varies from lift to lift. One can max more often on the bench press than exercises which actively use legs.

    - Most of the >90% 1RM lifts are performed on the bottom of the top intensity zone.

- The *training max* is the heaviest weight one can lift without strong emotional excitation. Well spotted maximal attempts reduce the excitation and increase the training max.

  For elite weightlifters the training max is about 85-90% of the competition max, with heavier athletes on the left of this range and lighter ones on the right. For non-lifters the training max is likely to be relatively higher—which explains why many strength programs for T&F athletes feature "unrealistic" loads like frequent doubles with 95% 1RM.

  The heart rate before the set can be monitored to identify the maximal training weight; if it goes up before the set, the weight is too heavy. The heaviest weight that does not elevate the athlete's HR beforehand is his training max on the given day.

  Unlike the competition max, the training max can be lifted in every training session; up to 6 lifts may be performed with the top weight—although this does not fit many training systems.

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Estimated 1RM (E1RM)

(Zonin, inspired by Chernyak)

The web and industry publications provide countless formulas to estimate the 1RM according to a given RM. Most of them are not reliable as a given RM corresponds to a % 1RM which varies considerably from subject to subject and lift to lift. E.g., with 80% 1RM there are athletes who can complete only 3 reps and others 22 and ever more. (Belsky). This variability increases the further one moves away from 1RM.

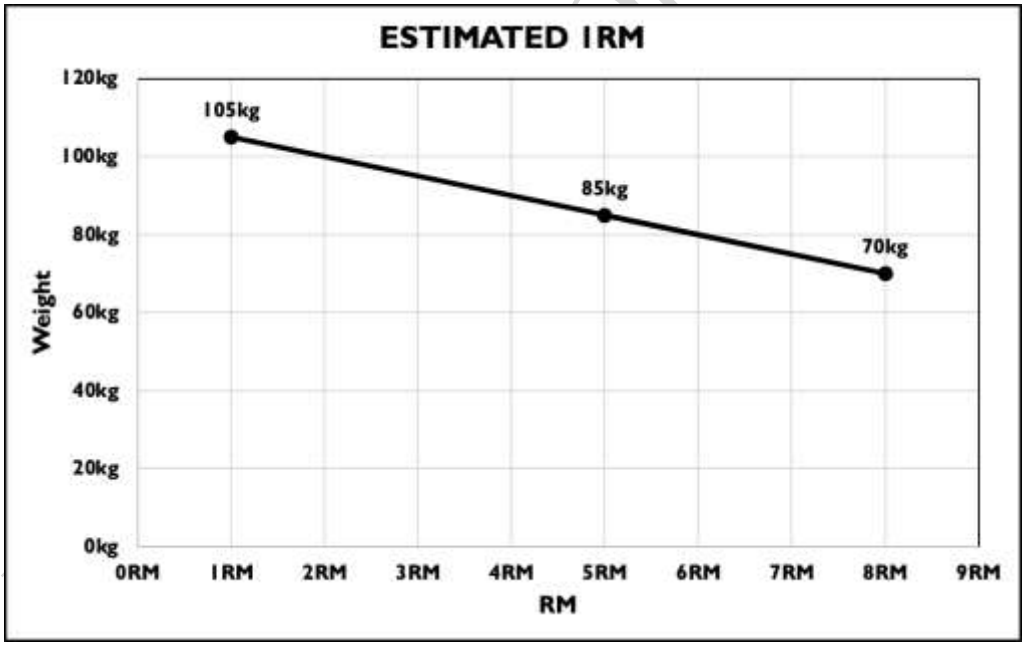
In general, it is nearly impossible to guesstimate the 1RM based solely on nRM data. (nRM—a repetition maximum with the number of reps "n." E.g., n=1, n=5, etc.) But if two different nRMs are known, where n is within a 3-10 range, things change.

At the basis of the following reasoning are the first two Euclidean postulates:

1. Between any two points it is possible to draw one and only one straight line.
2. You can extend a segment beyond the two points indefinitely.

If you know two different nRMs, you can plot everything on a graph and draw a straight line between the two points. The point where the drawn line meets the vertical axis passing through n=1 will give you the estimated value of 1RM. E.g.:

5RM = 85kg
8RM = 70kg
1RM = 105kg



Although the relationship between the weight and the RM is not actually linear, in a small RM range close to 1RM, it is *de facto* linear.

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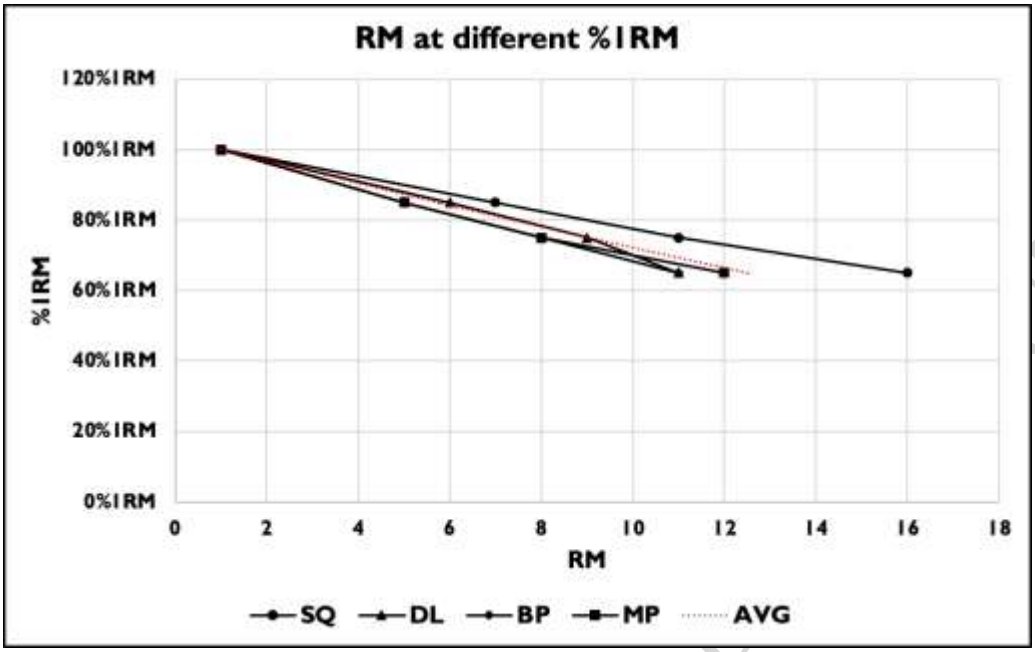
The graph below represents the average RMs at different % 1RM of my students (n=15) in different lifts (SQ, DL, BP, MP). As you can see, in the 75-100%1RM range the relationship is linear.



The E1RM formula:

$$E1RM = RM1 + \left[(n1 - 1) \times \frac{RM1 - RM2}{n2 - n1} \right]$$

E1RM—estimated 1RM
RM1—the first RM (<u>must be in a 3-5 range</u>)
n1—the number of reps in RM1
RM2—the second RM (<u>must be in an 8-10 range)</u>
n2—the number of reps in RM2

E.g.:

5RM = 85kg (RM1 = 85; n1 = 5)
8RM = 70kg (RM2 = 70; n2 = 8)

n1-1=5-1=4
RM1-RM2=85-70=15
n2-n1=8-5=3

85+(4x (15:3)) =105. The E1RM is 105kg.

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Estimating repetition maximums (RM) at different % 1RM: methods and applications

(Zonin)

Introduction

To optimize strength training, in some cases it is useful to estimate the maximum number of repetitions that can be performed at a given percentage of 1RM ("P%").

The best approach is to start from a test performed with a load equal to 80% of 1RM, which allows for a reliable estimation of the decrease in repetitions as the percentage of load changes.

However, if a load corresponding to exactly 80% of 1RM is not available, we can use a test performed at a slightly different percentage and apply an adaptation formula to obtain equally accurate estimates.

Why is 80% of 1RM a reliable reference?

The 80% of 1RM load represents a balance between two key factors:

1. Linearity of the load-RM relationship: Between 80% and 100% of 1RM, the relationship between load and the maximum number of repetitions is nearly linear, making estimates more reliable.

2. Good variability in repetitions: 80% of 1RM is light enough to allow for a good range of repetitions between individuals and even between different lifts of the same individual. At the same time, it is heavy enough to provide a solid reference for estimating RM at other load percentages.

Case 1: The test is performed exactly at 80% 1RM (or between 77.5% and 82.5% 1RM)

We can estimate the maximum number of repetitions at any other percentage of 1RM between 70% and 99%, using the following formula:

$$RM(P\%) = 1 + \frac{(n-1)}{20} \cdot (100 - P)$$

Where:

- RM(P%) = estimated maximum repetitions at percentage P% of 1RM.
- n = maximum number of repetitions completed at 80% of 1RM.
- P = percentage of 1RM for which we want to estimate the maximum repetitions (P% × 100).

If the maximum repetitions test is performed with a load between 77.5% and 82.5% of 1RM, using this formula still provides reliable results, with a maximum error of ±1 repetition compared to the more accurate formula.

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Example

Suppose an athlete can complete 8 repetitions at 80% of 1RM. We want to estimate how many repetitions they could complete at 70% of 1RM. Applying the formula:

RM (70%) = 1 + ((8 - 1) / 20) x (100 - 70)
RM (70%) = 1 + (7/20) x 30
RM (70%) = 1 + 10.5 = 11.5

Rounding down: RM (70%) ≈ 11 reps. Thus, the athlete could complete about 11 reps with 70% of 1RM.

To save time, if your RM test was between 77.5% and 82.5% of 1RM, you can use the table below to estimate RMs at different percentages of 1RM.

%1RM	RM@80%1RM										
	4	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
100%	1	1	1	1	1	1	1	1	1	1	1
99%	1	1	1	1	1	1	1	1	1	1	1
98%	1	1	1	1	1	1	1	2	2	2	2
97%	1	1	1	1	2	2	2	2	2	2	2
96%	1	1	2	2	2	2	2	3	3	3	3
95%	1	2	2	2	2	3	3	3	3	4	4
94%	1	2	2	2	3	3	3	4	4	4	4
93%	2	2	2	3	3	3	4	4	4	5	5
92%	2	2	3	3	3	4	4	5	5	5	6
91%	2	2	3	3	4	4	5	5	5	6	6
90%	2	3	3	4	4	5	5	6	6	7	7
89%	2	3	3	4	4	5	5	6	7	7	8
88%	2	3	4	4	5	5	6	7	7	8	8
87%	2	3	4	4	5	6	6	7	8	8	9
86%	3	3	4	5	5	6	7	8	8	9	10
85%	3	4	4	5	6	7	7	8	9	10	10
84%	3	4	5	5	6	7	8	9	9	10	11
83%	3	4	5	6	6	7	8	9	10	11	12
82%	3	4	5	6	7	8	9	10	10	11	12
81%	3	4	5	6	7	8	9	10	11	12	13
80%	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
79%	4	5	6	7	8	9	10	11	12	13	14
78%	4	5	6	7	8	9	10	12	13	14	15
77%	4	5	6	7	9	10	11	12	13	14	15
76%	4	5	7	8	9	10	11	13	14	15	16
75%	4	6	7	8	9	11	12	13	14	16	17
74%	4	6	7	8	10	11	12	14	15	16	17
73%	5	6	7	9	10	11	13	14	15	17	18
72%	5	6	8	9	10	12	13	15	16	17	19
71%	5	6	8	9	11	12	14	15	16	18	19
70%	5	7	8	10	11	13	14	16	17	19	20

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Case 2: The test is not performed exactly at 80% 1RM (and falls outside 77.5-82.5% 1RM)

If the athlete cannot test with a load equal to 80% of 1RM but must use a slightly different load outside the 77.5% - 82.5% range, we can still estimate the maximum repetitions at any other percentage of 1RM by adjusting the calculation. The formula to use is:

$$RM(P\%) = RM(P_t\%) + \frac{RM(P_t\%) - 1}{100 - P_t} \cdot (P_t - P)$$

Where:

- RM(P%) = estimated maximum repetitions at percentage P% of 1RM.
- RM(Pt%) = maximum number of repetitions completed with the tested load (Pt%).
- P = percentage of 1RM for which we want to estimate the maximum repetitions (P% × 100).
- Pt = actual percentage of 1RM used during the test (Pt% × 100).
- 1RM = maximum load that can be lifted for a single repetition.

Example

An athlete's 1RM is 120 kg. He cannot test his RM at 80% 1RM (96kg) but he can with 100kg, which corresponds to 83.3% 1RM. The athlete completes 6 repetitions with 100kg. We want to estimate how many repetitions he could perform at 75% 1RM (90kg). Applying the formula:

RM (75%) = 6 + ((6 - 1) / (100 - 83.3)) x (83.3 - 75)
RM (75%) = 6 + (5 / 16.7) x 8.3
RM (75%) = 6 + 2.49
RM (75%) = 8.49

We round down: RM (75%) ≈ 8 repetitions.

Why do we round down?

Because an athlete cannot perform a fractional number of repetitions in a single set, e.g., 8.9 repetitions.

Conclusions

- If the tested load is exactly 80% 1RM or falls between 77.5% and 82.5%, use the first formula to estimate RM at any other percentage, with a maximum error margin of ±1 repetition.
- If the tested load is < 77.5% or > 82.5% 1RM, it is preferable to use the second formula for a more precise estimation.
- Both formulas assume a linear relationship between load and RM, which remains reliable within the 70-99% 1RM range.
- Rounding down ensures consistency in results, as athletes can only complete whole numbers of repetitions.

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5. Reps and effort

- 1-6 is the optimal repetition range for strength.
- Minimize the share of singles and doubles.
- If hypertrophy is undesirable, favor triples and fours; if desirable, fives and sixes.
- Do not confuse intensity with effort or *rate of perceived exertion*.
- RPE is a subjective measure of effort, frequently measured on a scale from 1 to 10.
- Related to the RPE, a "buffer" or RIR (reps in reserve) indicates the difference between the maximum number of repetitions that could be executed in a row and those that are performed in a set with a given weight. E.g., in a set of 6 reps with a 10RM weight the buffer is 4 reps (or 4 RIR).

  In Plan Strong™ and some other systems the number of reps per set may be prescribed relative to the RM, e.g., 70% 1RM x 2/3 RM, 80% 1RM x 1/2 RM, etc.

  RIR allows to structure training sessions around a specific proximity to RM. It is also possible to design training cycles based on the manipulation of RIR, increasing either the weight or the reps each week or session.

Sample cycle #1 based on RIR
(Weight remains the same, volume changes)

Week	Weight x Reps/Sets	RIR
1	5RM x 2/3	3
2	5RM x 3/3	2
3	5RM x 4/3	1
4	5RM x 5/3 (PR!)	0

On week 1 the lifter does 2/3 with his 5RM, so at RIR=3. The number of reps per set increases every week, and the RIR decrease accordingly. On week 1 the lifter would not have been able to complete 5/3 with his 5RM but, by the time he reached week 4, he got stronger and completed all sets of 5, achieving a PR. Two more self-explanatory examples:

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Sample cycle #2 based on RIR
(Weight changes, volume remains the same)

Week	Weight x Reps/Sets	RIR
1	75% 5RM x 5/5	~5
2	80% 5RM x 5/5	~4
3	85% 5RM x 5/5	~3
4	90% 5RM x 5/5	~2
5	95% 5RM x 5/5	~1
6	5RM x 5/5 (PR!)	0

Sample cycle #3 based on RIR
(StrongFirst, based on Ware;
both weight and volume change)

Week	Weight x Reps/Sets	RIR
1	80% 5RM x 5/3	~4
2	85% 5RM x 5/3	~3
3	90% 5RM x 5/2	~2
4	95% 5RM x 5/2	~1
5	5RM x 5	0
6	105-110% 5RM x 5 (PR!)	

- Mike Tuchscherer adapted Borg's RPE scale to powerlifting and tied the RPE to the RIR. At StrongFirst, in addition to the RIR, we also consider the drop in rep speed:

RPE Table (Tuchscherer, modified by Zonin)	
RPE	**Definition**
10	RIR=0—maximal effort
9.5	RIR=0—weight can still be lightly increased
9	RIR=1—challenging
8.5	RIR=1-2—last few reps done slowed down
8	RIR=2—last 1-2 reps done slowed down
7.5	RIR=2-3—last rep done slowed down
7	RIR=3-4—speed maintained with maximal effort
5-6	RIR=4-6—speed maintained with moderate effort
1-4	Very light to light effort

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In strength training most of the volume of work is done at a medium effort, in the 6-8 RPE range.

Only 5-10% of the total volume is executed in the 8-9 RPE range and 9-10 RPE are usually used only for testing 1RMs and attempting PRs, which is something that happens no more than 2-3 times per year.

The 5-6 RPE range is used mostly for light/active recovery days and the 1-4 RPE range for the warm-up and ramp-up.

RPE driven sample 8-week linear cycle

Week	Weight x Reps/Sets	RPE
1	69% 1RM x 5/5	6
2	73% 1RM x 5/5	7
3	77% 1RM x 5/5	7.5
4	81% 1RM x 5/5	8.5
5	85% 1RM x 5/5 (PR!)	9
6	89% 1RM x 3/3	7.5
7	93% 1RM x 2/2	8
8	97% 1RM x 1 101% 1RM x 1 105% 1RM x 1 (PR!)	8.5 9.5 10

In the above example, even though the weight keeps increasing steadily every week, the RPE drops on week 6, when the lifter cuts his reps and sets. The RPE drop is amplified if the lifter adds gear (e.g., knee wraps, a belt, a lifting suit, a bench shirt, etc.). Many classic US powerlifting cycles from the 1980s included such a "stealth deload."

RPE/RIR[3] vs. % 1RM

A % 1RM allows a very precise selection of the training load and meticulous planning. However, it doesn't consider the psychophysical state of the athlete on a given day or period.

Performance is influenced by several environmental factors, such as nutrition, sleep, work related stress, family issues, etc., and this is particularly evident in non-professional athletes who must deal with what happens in their lives outside the gym.

Also, there are some exercises (e.g., the barbell row) that by nature do not lend themselves to a 1RM test. It is not possible to achieve an accurate 1RM in kettlebell exercises, due to the large jumps between sizes. E.g., an athlete who might be able to complete 3 reps with a 40kg kettlebell but cannot press 44kg for a single.

[3] Thanks to Jose Luís Cortina, StrongFirst Certified Team Leader, for providing this material.

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There are also cases in which a 1RM test is not advisable, for instance beginners who still need to solidify their skills, athletes who have just recovered from an injury, or those who haven't practiced a given lift for some time.

In these cases, % nRM can be used in place of % 1RM. "n" stands for the maximum number of reps one can complete with a given weight on a given lift. E.g., 3RM, 5RM, 6RM, etc.

Still, neither the % 1RM nor the % nRM consider the athlete's psychophysical state at the given time.

Planning an athlete's training based on RPE/RIR allows to adapt the load to the individual response.

Experienced athletes have impressive self-awareness and are able to predict their performance based on the perceived effort.

Does this mean thar RPE/RIR should be used only with advanced athletes?

Yes and no.

RPE/RIR is a great way to assess the starting weights in beginners' cycles. In these cases, great precision is not necessary, as it is all about finding a weight that allows to complete "X" reps with a moderate RPE.

With the exception the case above, RPE/RIR is a valuable tool for more advanced athletes and should be implemented in a gradual fashion to allow them to get accustomed to its use.

The first step to get accustomed to using RPE/RIR is following a plan based on % 1RM or % nRM and keep track of the ongoing of the RPE/RIR across the sessions. This will allow to make a connection between the plan's parameters (intensity, volume, and density) and the perceived exertion.

The next step may be a mixed approach, such as using % 1RM, % nRM on heavy days and base the light days on RPE/RIR, or using % 1RM, % nRM for the main lifts and RPE/RIR for specialized variety and/or assistance lifts.

Following, RPE/RIR can be used in the off-season and/or preparation mesocycles, while sticking to % 1RM for the peaking/competitive phase.

RPE/RIR is a great training option also for those who prefer the % 1RM approach, but are going through challenging stages of their life such as wedding, newborn child, divorce, job stress, etc.

- "To prevent overstrain of the nervous system it is undesirable to improve the numbers through excessive willpower… It is advisable to set up ["stop signs"]…limit the motivation level…" (Kozhourkin)

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StrongFirst "Stop Signs" for strength training

- ✓ Rep speed slows down
- ✓ Tempo drops (pauses between reps—if applicable—increase)
- ✓ Technique changes in any way

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Relative Effort and Average Relative Effort (ARE)

Let's say an athlete needs to complete NL 20 at 75% 1RM (corresponding to his 10RM) in one session. There are several options; in the following example I am listing four:

1. *5/4 (total sets = 4)*
2. *2, 3, 5, 2, 3, 5 (total sets = 6)*
3. *3, 4, 6, 3, 4 (total sets = 5)*
4. *10, 6, 4 (total sets = 3)*

Of course, there are many other rep/set combinations, but let us focus on these above.

The average number of reps per set:

1. *5 —(5+5+5+5): 4 sets*
2. *3.33 —(2+3+5+2+3+5): 6 sets*
3. *4 —(3+4+6+3+4): 5 sets*
4. *6.67 —(10+6+4): 3 sets*

If we divide the numbers obtained above by 10 (as a reminder, the athlete is working with his 10RM) and transform them into % we will obtain the ***Relative Effort*** (RE):

1. *RE 50%*
2. *RE 33.3%*
3. *RE 40%*
4. *RE 67.7%*

This metric can be used to monitor how the effort varies from session to session and from week to week within a program.

What we have just seen was a simple case in which the athlete trains the lift using only one weight. Let us review an example with multiple intensities in one session:

- *70% 1RM (12RM): NL 10*
- *75% 1RM (10RM): NL 10*
- *80% 1RM (8RM): NL 8*
- *85% 1RM (6RM): NL 6*

Two examples of how the workload may be accomplished:

1. *70% x 5, 75% x 5, 80% x 4, 85% x 3/2, 80% x 4, 75% x 5, 70% x 5*
2. *70% x 5, 75% x 3, 80% x 2, 85% x 2/3, 80% x 2, 4, 75% x 3, 4, 70% x 5*

The average number of reps per set would be:

1. *5 @ 70%, 5 @ 75%, 4 @ 80%, 3 @ 85%*
2. *5 @ 70%, 3.33 @ 75%, 2.67 @ 80%, 2 @ 85%*

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The relative values of RE turn out to be:

1. *@ 70% 1RM, RE 50%; @ 75% 1RM, RE 50%; @ 80% 1RM, RE 50%; @ 85% 1RM, RE 50%*
2. *@ 70% 1RM, RE 50%; @ 75% 1RM, RE 33.3%; @ 80%, 1RM 33.4%; @ 85% 1RM, RE 33.3%*

Consequently, the ***Average Relative Effort*** (ARE) values are:

1. *ARE 50% — (50% + 50% + 50% + 50%) : 4*
2. *ARE 37.5% — (50% + 33.3% + 33.4% + 33.3%) : 4*

Same volume, same ARI, but totally different ARE!

Proposed optimal ARE (Zonin)	
Goal	**ARE**
Power	30-50%
1RM strength	40-60%
Strength and hypertrophy	50-70%
Classic bodybuilding	> 80%

RIR training vs. training to failure

In bodybuilding it is a common practice to push sets until concentric failure. This practice is also called training to "temporary muscular incapacity" (TMI). In bodybuilding it is common belief that only the very last reps of a set, the most strenuous ones, are those that lead to results.

In strength training, on the other hand, it is common to train with a buffer, to terminate a set before reaching the TMI.

- Some prescribe interrupting the set before the technique deteriorates.
- Others coach to stop the set as soon as the rep speed slows down.
- Soviet weightlifting coaches used to prescribe a repetition range from ⅓ to ⅔ of the RM at a given % 1RM.

Example 1—Sets pushed to failure

- *Bench press*
- *6RM = 100Kg*

Potential number of reps completed pushing each set to TMI:

- *6, 4, 2, 1*
- *NL 13*
- *High quality reps <50%*
- *The athlete will most likely be sore in the days that follow the session, with no chances to train the same lift for days.*

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Example 2—Sets with a buffer

- *Bench press*
- *6RM = 100Kg*

No. of sets and reps completed at 50% RM:

- *3, 3, 3, 3, 3, 3,…—and maybe more*
- *NL 18—and maybe more*
- *High quality reps—100%*
- *The athlete will not be sore in the days that follow and will be able to train the same lift several days in a row.*

The tables below show the Plan Strong™ reps per set guidelines.

Reps per set for lifters with average to high endurance with a given RM						
% 1RM	70%	75%	80%	85%	90%	95%
RM, ≥	12RM	10RM	8RM	6RM	4RM	≥2RM
Optimal	6	5	4	3	2	1
Range	4-8	3-6	3-5	2-4	1-2	1
Warm-ups; comp week	2-3	1-2		1		

Reps per set for lifters with low endurance with a given RM						
% 1RM	70%	75%	80%	85%	90%	95%
RM	10RM	8RM	6RM	5RM	3RM	2RM
Optimal	5	4	3	2	1-2	1
Range	3-6	3-5	2-4	2-3	1-2	1
Warm-ups; comp week	1-3	1-2		1		

Other Plan Strong™ guidelines:

- Vary as much as possible the number of reps from one set to another within the given range.
- Ladders are a very effective tool to manage high volumes while managing neural and metabolic fatigue.

Ladders vs. static reps

Example 1—Constant reps

- *Bench press*
- *75% 1RM (10RM)*
- *NL 25*
- *Rep range: 3-6*
- *5 sets of 5 reps—the well-known 5x5*

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Example 2—Ladder

- *Bench press*
- *75% 1RM (10RM)*
- *NL 25*
- *Rep range: 3-6*
- *3, 4, 6, 3, 4, 5*
  or
  2, 3, 5, 2, 3, 5, 2, 3

Some advantages of ladders:

- More sets equal more set-ups and more skill practice.
- Same intensity, lower RPE in most sets allows to manage higher volumes with better technique. *E.g., 5/5 (NL 25) vs. 2, 3, 5, 2, 3, 5, 2, 3, 5 (NL 30). In the latter case, the volume is higher, but only 3 out of 5 sets are at a high RPE.*

How to set up a ladder according to the Plan Strong™ guidelines

1. Test your RM.
2. Calculate 1/3 of the RM and round it to the nearest whole number.
3. Calculate 1/2 of the RM and round it to the nearest whole number.
4. Calculate 2/3 of the RM and round it to the nearest whole number.

The three numbers you obtained, from the smallest to the largest, are the repetitions of your ladder.

RM	Ladder
4	1, 2, 3
5	2, 3, 3
6	2, 3, 4
7	2, 4, 5
8	3, 4, 5
9	3, 5, 6
10	3, 5, 7
11	4, 6, 7
12	4, 6, 8
13	4, 7, 9
14	5, 7, 9
15	5, 8, 10

Fibonacci ladder

In this particular type of ladder, the number of reps of the third set is equal to the sum of the number of reps of the first set and the number of reps of the second set—a Fibonacci numbers ladder.

While the number of reps of the third set is always equal to 2/3 RM, the number of reps of the first and second sets could be less than 1/3 RM and 1/2 RM, respectively.

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This leads to lower ARE values and better fatigue management, with the consequent ability of the athlete to tolerate higher volumes of work than with the traditional ladder in which all rep jumps are the same.

Fibonacci Ladder	
RM	**Ladder**
4	1, 2, 3
5	1, 2, 3
6	1, 3, 4
7	2, 3, 5
8	2, 3, 5
9	2, 4, 6
10	3, 4, 7
11	3, 4, 7
12	3, 5, 8
13	4, 5, 9
14	4, 5, 9
15	4, 6, 10

Ladders and ARE

You may select and manage ladders according to the target ARE and your training goal. For instance, for higher AREs you may skip the first set of a ladder.

RM	Standard Ladder		Fibonacci Ladder	
4	1, 2, 3	2, 3	1, 2, 3	2, 3
5	2, 3, 3	3, 3	1, 2, 3	2, 3
6	2, 3, 4	3, 4	1, 3, 4	3, 4
7	2, 4, 5	4, 5	2, 3, 5	3, 5
8	3, 4, 5	4, 5	2, 3, 5	3, 5
9	3, 5, 6	5, 6	2, 4, 6	4, 6
10	3, 5, 7	5, 7	3, 4, 7	4, 7
11	4, 6, 7	6, 7	3, 4, 7	4, 7
12	4, 6, 8	6, 8	3, 5, 8	5, 8
13	4, 7, 9	7, 9	4, 5, 9	5, 9
14	5, 7, 9	7, 9	4, 5, 9	5, 9
15	5, 8, 10	8, 10	4, 6, 10	6, 10
	ARE≅50%	ARE≅60%	ARE≦50%	ARE≅50%

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6. Rest

a) Frequency

- The following table refers to all training days, not just developmental (heavy) ones.

1 day/week →	6 days/week
Time efficient	Time consuming
Little skill practice	Plenty of skill practice
Slow strength gains	Rapid strength gains
	Fragmentation of a given volume into more sessions improves results—but may not always fit the progressive overload model.
	Shortening the intervals between training sessions allows a better summation of the training effects
Slow detraining	Rapid detraining
Possible conversion of IIA myosin to IIX	Likely conversion of IIX myosin to IIA
No improvement in CP and glycogen capacity	Increased CP and glycogen capacity
More soreness	Less soreness
Simple planning	Precise planning and monitoring

b) Rest between repetitions, sets, and series

Different types of rest intervals within a session			
	Ordinary	**Stress**	**Stimulation**
Description (Matveev)	"Relative normalization of the function." By its end, the work capacity approaches the level before the previous exercise bout, to the point where neither the quality nor the quantity suffers.	"Its duration is so short that the next load [set] is overlaid onto the remaining functional activity of certain systems of the organism caused by the previous load [set]. As a result, the effect of the next load [set] is increased." Performance does not have to go down but it comes with a greater effort.	The shortest interval after which the performance increases. Facilitation stops happening as fatigue sets in.
	A given time interval may change with fatigue from stimulation to ordinary to stress.		
Uses	Most common in strength training	With the exception of the rest-pause technique, rarely used in strength training	Standard in classic US powerlifting cycling
		Common in hypertrophy training	Preferred for "greasing the groove"

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- Performing reps non-stop shifts the metabolism towards glycolysis, while introducing brief intra-set pauses enables greater reliance on the CP mechanism. E.g., a 5sec rest between reps of a 10-rep set reduces the lactic acid concentration by 50%.

- The recovery of at least three functions must be considered**: CP**, **pH**, and **CNS** excitability. The first recovers rapidly, the second slowly, the third is in the middle.

- **1/2/3 rule:** ~2/3 of recovery takes place in the first 1/3 of total recovery duration, ~30% in the second, and ~5% in the third.

- *Heterochronicity* of functions' recovery and the exponential decay recovery dynamics apply to different time scales: a set, a day, a week…

Approximate recovery timeline for strength exercises	
Time	**Events**
5sec	Lactic acid reduced by 50% compared to doing reps consecutively
30sec	CP ~50% recovered
1min	Steepest CP recovery
2min	CNS excitability starts declining
3min	Fast CP recovery phase; almost complete CP recovery
>3min	Slow CP recovery phase
5min	CNS excitation rapidly declines after this point
5-8min	CP fully recovered
10min	Performance fully recovers after a set to RM
20min	20% performance supercompensation after a set to RM
15-20min	CP supercompensation peak after strength exercise to RM
30-90min	Total lactic acid clearance

- If the day's load for a given exercise is greater than NL 30 or it takes longer than 30min, divide the load into multiple series separated by unrelated exercises or ≥30min of passive rest.

c) Nature of rest

✓ Active rest: walk

✓ Sechenov's principle

✓ "Fast & loose"

✓ Self-massage, stretching, and low intensity exercise between sets accelerate the recovery by 10-15%.

d) Training session duration

- As the training session progresses, the damage by free radicals intensifies, reaching its maximum towards the end of the session due to cell membranes' damage. Short sessions are preferable.

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7. Exercise dynamics

- Do not confuse the bar speed with the tempo (cadence).
- At StrongFirst we classify the concentric lifting speed as follows:
  - "Fast"—as fast as possible without jerking. Use compensatory acceleration.
  - "Medium"—a comfortable, confident speed, corresponding to a first attempt in a powerlifting competition.
  - "Slow"—the speed of a hard-fought PR single.
- Between fast, medium, and slow speed, the most effective one is medium (greater gains and slower accommodation).

  Varying the speeds is most effective. It can be done within a microcycle or in separate mesocycles.
- Eccentric phase options:
  - Controlled
  - "Dive bomb"
  - Combination of the above
  - No negative (dropping the bar)
- Eccentric to concentric transition options:
  - Dead start (no eccentric)
  - Touch and go (several variations)
  - Pause with tension
  - Relaxed pause
- Isometrics produce rapid results and rapid accommodation (6-8 weeks).
- The *static-dynamic method* (introducing pauses into a dynamic repetition) is highly effective.

8. Preceding load

- The closer strength and endurance sessions are in time, the more they interfere with each other. It is better to train the two on alternate days rather than on the same day.
- You must be fresh coming into a neural strength training session.
- A low volume, medium to high intensity session facilitates the next day's load.
- A low to medium volume, high intensity session for the upper body facilitates the lower body the day after and vice versa.

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- If on the first day one of the load variables, volume or intensity, is high and the other is medium, the second day's training will be conducted in a state of fatigue. The training effects of both sessions will be summated. It is an effective advanced tactic, rarely used in traditional progressive overload, but common in variable overload.

- A high volume lower body session on day one will have a negative effect on upper body performance on day two. However, if the upper body session is hypertrophy oriented, it will benefit from the anabolic hormones released by the lower body session.

9. Following load

- "…When learning to cook a new dish from a recipe, an inexpert cook catches mostly the quantity of the ingredients; an expert cook—rather the way and the sequence of their addition." (Verkhoshansky)

- The recovery time after two adjacent training sessions with significant loads but different foci is insignificantly longer compared to just the first session.

- Deloading days after high load days are especially effective if they employ exercises of totally different nature from the ones used on the high load days.

  However, a low volume, light to medium intensity session the day after a large load has a strong anabolic effect.

- Avoid endurance training for 24-36 hours after a hypertrophy training session.

- "Artificial acceleration of the recovery process may compromise the natural course of long term adaptation to strenuous muscular activity and thus can be justified only in extreme cases of heavy excessive fatigue…frequent and regular use of non-specific recovery means is in biological contradiction with application of large training goals. An increase of the volume and intensity of the latter aims to disrupt homeostasis as the main condition of the organism's adaptation to loads and to train the recovery processes. An intervention into the course of the recovery upsets the organism's ability to perfect this process naturally." (Verkhoshansky)

10. Load dynamics

- On one hand, strength grows faster with a constant weight because it is a "definite stressor." (Vorobyev)

  On the other, lifting variable weights is less tiring and improves extrapolation of adaptation to other weights.

- The RPE and the repetitions per set may also change within a session.

- Load dynamics on the micro, meso, and macro scales discussed elsewhere in the document.

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Trainable strength factors and corresponding methods: morphology

Adaptations	Methods	Notes
Myofibrillar hypertrophy	Increase volume with 70-80% 1RM, 1/2-2/3 RM, and ordinary rest periods (~3min)	There are many methods; we are only listing the top choices at StrongFirst.
	Interrupted sets: sets of 5 reps with 8RM with 20-30sec of rest between them, 2-3 series	
	Multiple back-off sets 5-10 reps with 50-65% 1RM and ordinary rest periods (~3min)	
	1-3 back-off sets of 8-10 reps, RPE 8-8.5	
	5-10 sets to RM with 8-10RM with 5-10min rest periods	
Sarcoplasmic hypertrophy	Traditional bodybuilding	Increases strength only for heavyweights through improved tissue leverage Not recommended to athletes whose sports require high levels of relative strength Not recommended to athletes whose sports demand high speed due to increased intramuscular friction
Conversion of IIA myosin to IIX	Extremely low volume and frequency taper	Applicable mostly to powerlifters and throwers
Connective tissue hypertrophy	Very high volume of controlled full ROM movements with minimal resistance	Recommended early in the athletic career and in the off-season
	Supramaximal methods	

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Trainable strength factors and corresponding methods: neurology

Adaptations		Methods	Notes
Increased neural drive		**Near-maximal, maximal, supramaximal, dynamic efforts:**	
		Lifts in the 91-100% 1RM zone	High risk of overtraining
		Lifts with 80-90% to RM	To be done infrequently
		Artificial controlling environment (professionally done forced reps)	For experienced athletes' peaking
		Supramaximal eccentrics	
		Compensatory acceleration training with 60-80% 1RM	Less stressful than heavy lifting; enables to reduce the total volume
		Isometrics	Rapid strength gains and short residuals
		Impact method ("plyometrics")	An advanced method
Sensitization of motor neurons to neural drive	Synaptic potentiation (perforated synapse)	Same methods as listed above for increasing the neural drive	
	Synaptic facilitation, myelination	High volume specific practice with 70-85% 1RM and 1/3-2/3 RM (Grease the Groove, Plan Strong™, Strength Aerobics)	GTG and A+A can be done on light and medium days
		Visualization	Psych (Biasiotto)
Intermuscular coordination		Treating strength training as a skill practice	
		Specialized variety	
Disinhibition		Avoiding muscle failure	The synaptic weight turns negative when an attempted action is not completed as intended.
		Sports psychology techniques	Psych (Biasiotto)
		Pretension	
		Successive induction (active negative)	
		Same methods as listed above for increasing the neural drive	
		"Dive bomb" full ROM lifts	Risky advanced methods (Hatfield)
		Tendon jerks: yanking on a loaded DL bar	
		Training methods for spinal stability	Both morphological and neurological; taught in the StrongFirst curricula

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Multi-year training

- The range of effective training means and methods narrows as the athlete advances. In the early years of an athletic career, it is best to minimize the use of advanced means and methods to keep the athlete more responsive to them later in his career. (Platonov)

- Avoid excessive variety for beginner and intermediate athletes. "Constantly renewing the means and methods slows down the motor qualities' development, since this [novelty] is an excessively hard trial for an organism unprepared for such strong and diverse stimuli." (Valmeyer)

- In young athletes, high loads, highly specialized exercises, and means of accelerated restoration rapidly lead to a premature slowdown of progress and minimize the effective training arsenal at the peak of the athletes' careers. (Platonov)

- The quicker an adaptation is formed, the quicker it is lost. Applied to long term training, the are many examples of elite athletes making a sudden two to three-fold load increase within a year, quickly reaching extraordinary performances—and quickly leaving the sports, unable to sustain them. At the same time, athletes who gradually increase their loads[4] over the years tend to reach high levels slower but are also able to compete longer at the elite level. (Platonov)

- Older and more experienced athletes improve at a slower rate but also detrain at a slower rate. Their training volume is 20-25% lower than younger athletes'.

Macrocycle

- "The major two tasks of all macrocycles are the creation of movement potential and its realization." (Viru)

- The length of a *macrocycle* ranges from several months to 4 years.

  In the US powerlifting cycling model, a macrocycle is 8-16 weeks long and is made up of logically sequenced mesocycles leading up to a major competition.

  In weightlifting a macrocycle (traditionally called the "big cycle") is typically made up of three periods, *preparatory* (1-2 months), *competition* (1 month), and *transition* (1 week to 1 month).

  The preparatory period builds a "foundation." The volume is high and the intensity increases slowly.

  "The goal of the competition period is to reach the highest athletic form and to realize it." A typical length is one month. "The competition period is characterized by a lower volume of loads and their maximal intensity." (Roman)

- "Division of the annual training into many [macro]cycles is not sensible. In that case, to prepare for every meet and to post a certain result the athlete must taper for it, that is sharply lower the volume of

[4] Note that this applies to a long-term trend; not short-term variability.

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the training load every time. As there are many cycles this decrease takes place often and the athlete does not get a chance to perform the necessary volume of training and thus build the functional base for increasing the results... Usually division of annual training into a high number of short cycles and the desire of the athlete to do his best in every competition lead to a plateau and even a decrease of the results." (Roman)

Training vs. peaking

- "Training" in the preparatory period develops long term adaptations (morphological, neurological, and, to a lesser degree, biochemical).

  "Peaking" in the competition period mobilizes the organism for a short-term performance spike by:

  1) **Increasing the intensity to >90% 1RM**

  Heavy weights increase the CNS excitability or "tonus" by pushing nerve cells from the *normal* mode to the *forced* mode. (If they stay in this mode too long, they transition to the *pathological* mode, which is why an attempt to sustain a peak leads to a crash.)

  Experience shows that one can "spike" the CNS twice starting 14 days before the competition. Following is a sample peaking cycle.

TSC deadlift peaking cycle	
Week	**Load, % 1RM**
1	85% x 1, 80% x RM
2	88% x 1, 83% x RM
3	91% x 1, 86% x RM
4	94% x 1
5	**TSC:** Max

  2) **Sharply tapering the volume,** e.g., by 1/2 or 2/3

  At the very least, this improves performance by eliminating all traces of fatigue (*fitness-fatigue* model).

  At the most, if one practices *concentrated loading* (as opposed to *distributed loading* in traditional programming), it allows the athlete to realize *delayed training effects*.

  In addition, the myosin of some of the type IIA fibers converts to type IIX through *MHC IIX overshoot*.

  3) **Optimizing the timing of the final training sessions**

     a. In US powerlifting cycling reduced training frequency further facilitates the IIA to IIX myosin conversion.

     b. In Soviet weightlifting and Russian powerlifting, the rhythm of the loads (typically every other day plus a specific warm-up 24 hours before the meet) "leads" the organism to a peak.

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- A key point: in training, the load must increase at the same RPE. If it goes up but it takes a greater effort, it means one of the three things:

    - You are peaking.
    - You are purposefully *overreaching* with a plan to deload and realize delayed training effects later.
    - You do not know what you are doing.
    - You are having a bad day.

Mesocycle

- A *mesocycle* consists of 2-8 microcycles.

  A mesocycle has a particular purpose. Types of mesocycles:

    - *Introductory*
    - *Loading*
    - *Stress*
    - *Pre-competition*
    - *Competition*
    - *Restorative*

  A mesocycle ~ a training "block."

- A single loading mesocycle is sufficient to show improvement in the trained function. No improvement indicates poor planning.

- A complete adaptation takes at least 6 weeks. Research by Neumann led to the East German practice of planning elite athletes' training in 6-week units.

  "The adaptations of the human organism follow a regular pattern. During the first 1 to 2 weeks of a new training cycle the body adapts quickly to the new stimulus. In the next few weeks, the power of this same stimulus to provoke an adaptation will progressively fade to end completely after 6 weeks. We therefore call week 1 and 2 of the adaptation process the "fast adaptation phase" and weeks 3 to 6 the "stabilization phase."

  "Why wait 6 weeks before introducing a new stimulus or a new training load? Why not change the training plan after 2 weeks when the stimulus has induced its fast adaptation? The reason is that weeks 3 to 6 are necessary to stabilize the adaptations brought about in weeks 1 and 2." (Olbrecht)

- Since the strength improvement curve flattens out after several weeks, some specialists believe that strength training means and methods should pursue aggressive rates of adaptation and be changed after 2-3 weeks. However, the longer the training, the deeper the adaptation caused by it and the longer it remains in detraining. (Yampolskaya & Yakovlev)

  Gradualness of progression implies not only increasing loads according to the athlete's individual adaptive capacities but also staying with them long enough to assure adaptation.

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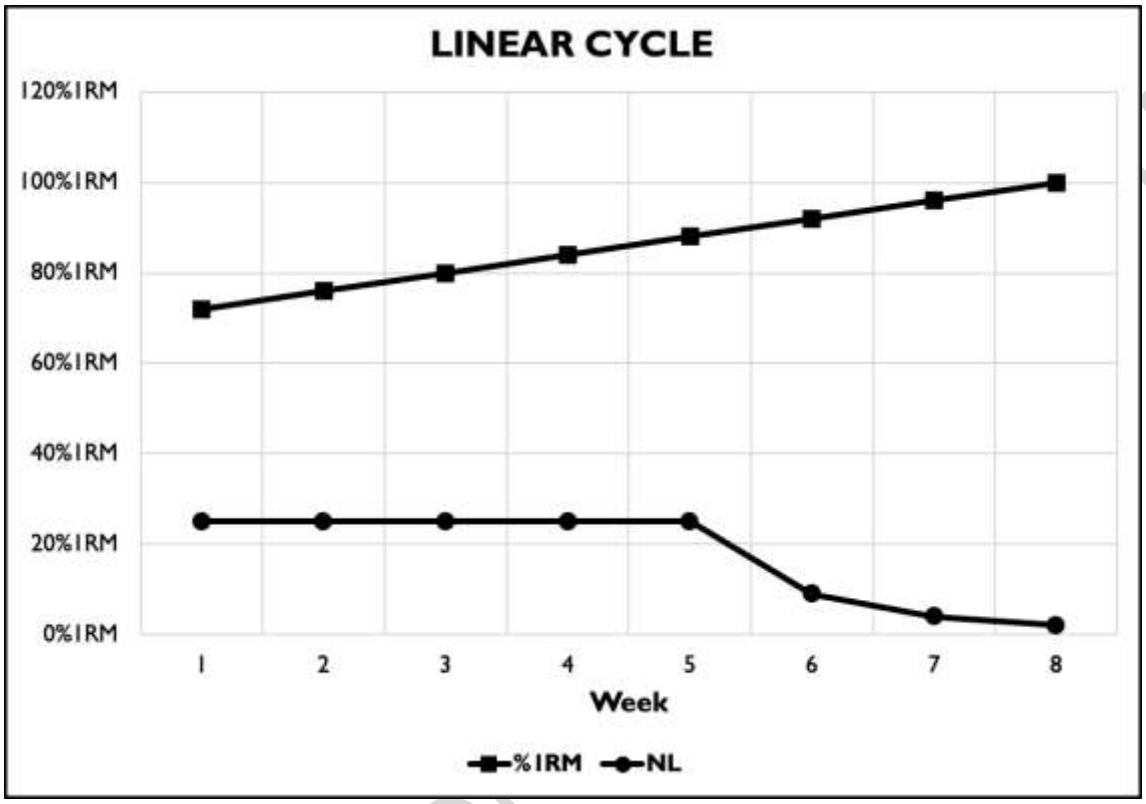
Linear cycling

- "The temporary loss of adaptivity has made it necessary to employ a cyclic structure in training." (Viru)
- Many US powerlifting cycles from the 1980s and the 1990s used the linear progression.



- Build 4-6 week mesocycles in such a manner that every consecutive main training session of the week is harder than the last week's, with the weight and reps peaking in the last week. This is a classic progressive overload template. E.g.,

Off-season 5RM cycle (StrongFirst, based on Ware)	
Week	**Load**
1	80% 5RM x 5/3
2	85% 5RM x 5/3
3	90% 5RM x 5/2
4	95% 5RM x 5/2
5	100% 5RM x 5
6	105-110% 5RM x 5

Once you have identified the set and rep scheme, set a reasonable RM or SM goal for the end of the mesocycle.

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Increase one load variable to build up (weight, the number of reps per set, the number of sets) and keep the rest constant. "Ideal training involves changing just one of many variables, until that variable reaches a constant. Then you change another, and then another until you reach your goal." (Morehouse)

Then work back from the projected PR, taking uniform steps, progressing in RPE from ~6-7 to ~8.5-9.

In a month with distributed loading, one should load the endocrine system for 2 weeks and unload for 2.[5] Adjust your load jumps accordingly.

E.g., in a 4-week mesocycle by Gallagher, the lifter sets a RM PR in week 4 and repeats his old PR in week 3. This makes for 2 easier and 2 hard weeks.

The two heavy weeks do not have to follow each other

Deadlift training template (Columbo): week 1—medium, week 2—heavy, week 3—medium, week 4—very heavy.

The same model can be applied to a structured cycle, e.g.:

"Ukrainian 4s" cycle (StrongFirst, based on Solovyev, Solovyev, Naleykin, Ivanenko)	
Week	**Load**
1	75% x 4/4
2	82.5% x 4
3	77,5% x 4/4
4	85% x 4
5	80% x 4/4
6	87.5% x 4
7	82.5% x 4/4
8	90% x 4

If the variable chosen for progression is intensity, progress in 2-5% 1RM increments. The exact value within that range depends on your strength endurance; see Reload and Reload 2.0.

Build up to a RM or SM PR with progressively harder weeks.

- Instead of planning 6-week blocks, one can set up an open ended cycle, to be terminated when a certain RPE is reached. Starting at 6-7 and finishing at 8.5-9 is common.

  E.g., the Power to the People! 2.0 cycle:

  - Start with 80-85% of 5RM.

[5] Unless the RPE never exceeds 8. In that case the endocrine system can handle weekly loading. The less you load, the less you have to unload.

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- Do a set of 5, then 3, and finally 2 with the same weight.
- Add 2-3% 1RM every session, until you hit your 5RM—adding more weight next time would be impossible or extremely challenging.
- Next session add weight as usual but do only a triple and a double.
- The following session add weight once again and do only a double.
- Next time test your 1RM.

Note how a foolproof peaking phase with lower reps is built in.

- If the intensity is constant and volume is the variable to progress, keep in mind that:
  - It is easier to add sets than to add reps. Thus, first add sets, then reps.

    E.g., an excerpt from the "Jurassic Rite of Passage" from Kettlebell 201 Workshop:

    1. (1, 2, 3) x 3 = 18
    2. (1, 2, 3) x 4 = 24
    3. (1, 2, 3) x 5 = 30
    4. (1, 2, 3) x 6 = 36
    5. (1, 2, 3) x 7 = 42
    6. (1, 2, 3) x 8 = 48
    7. (1, 2, 3) x 9 = 54
    8. (1, 2, 3) x 10 = 60
    9. (1, 2, 3, 4) x 3 = 30
    10. (1, 2, 3, 4) x 4 = 40
    11. …

  - When you add reps, it is preferable to start adding from the last set and work back towards the first set. E.g., the Fighter Pullup Program and plan 414.
  - When volume is the variable being increased, a mix of the three types of progressive overload (linear, wave, step) is common. E.g., a linear progression with periodic stabilizations at some level and an occasional back-off.

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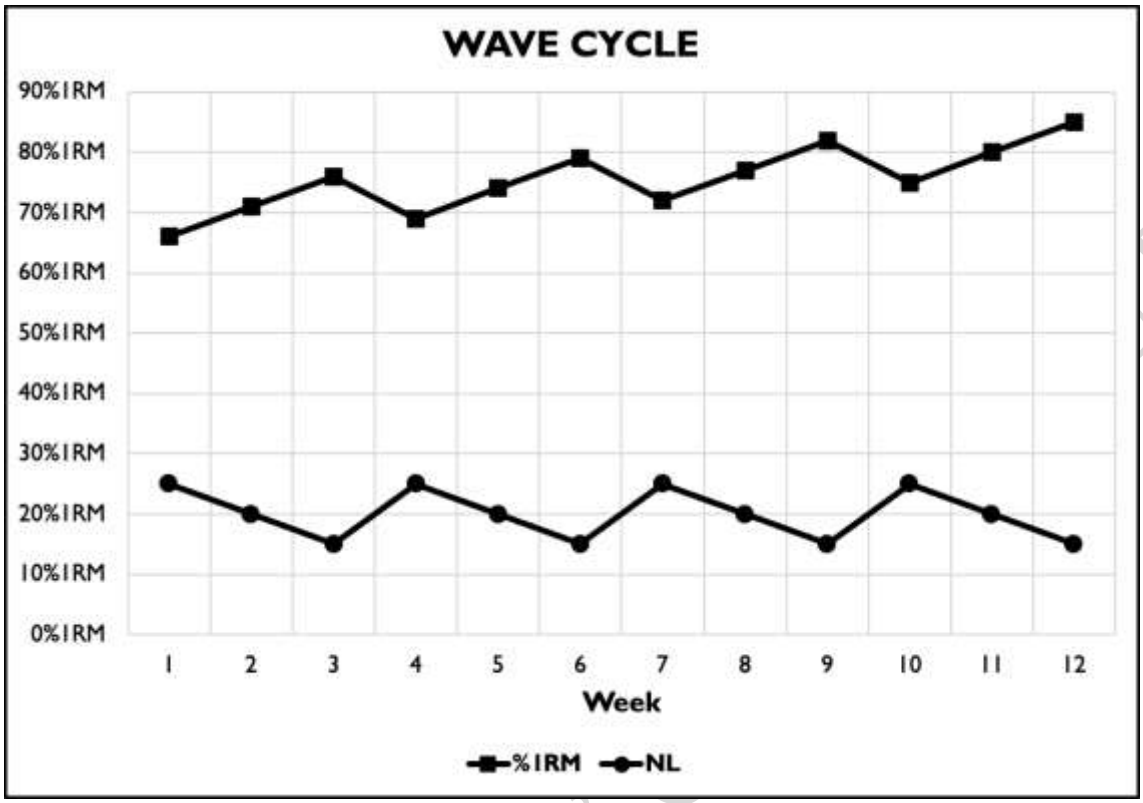
Wave cycling

- "…the preferred dynamic is wavelike…with a steep increase… It is not accidental that this type of load dynamics is often predominant in training of athletes making progress." (Matveev)



- The classic wave loading tactic is building a 6-week mesocycle made of 2 ascending waves, two steps forward and one step back. E.g., the "Fast Tens" progression for the kettlebell C&P:

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
2 x (5, 10)	3 x (5, 10)	4 x (5, 10)	3 x (5, 10)	4 x (5, 10)	5 x (5, 10)

- Some US powerlifting cycles from the 1980s and the 1990s used the wave progression in a big cycle. E.g., a cycle by Gaugler for a 500-pound deadlifter:

Week	*Load*
1	*390x3/2*
2	*410x3/2*
3	*430x3/2*
4	*450x3/2*
5	*405x3/2*
6	*425x3/2*
7	*445x3/2*
8	*465x3/2*

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- If a linear cycle that increases the load once a week and features light days, it uses a form of wave loading.

- In Soviet strength programming, especially in WL and T&F, both volume and intensity are frequently increased simultaneously, although at different rates, in the preparatory period.

  E.g., Dyachkov would increase the volume for 3-4 weeks and the intensity for 1-2 weeks before cycling back.

  Following are examples of such loading:

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
5/3	5/4	5/5	5/3	5/4	5/5
72% 1RM	74%	76%	78%	80%	82%

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
5/3	5/4	5/5	5/3	5/4	5/5
75%	80%	75%	80%	75%	80%

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
5/3	5/4	5/5	5/6	5/2	5/1
76%	78%	80%	76%	81%	86%

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
5/3	5/4	5/5	5/1	5/2	5/3
76%	78%	80%	82%	76%	84%

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
5/3	5/4	5/5	5/6	5/2	5/1
75%	78%	70%	78%	75%	88%

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Intensity	6RM					
NL	15	20	25	30	20	30
Reps/sets	3/5	3/7	4/8	3/10	4/5	5/6

This is not traditional progressive overload, since more than one variable is increased and the mesocycle does not necessarily end in a PR.

Such programming often falls under the concentrated loading category and/or variable loading.

The main benefit is the summation of high volume and high intensity for a strong adaptive response.

The downsides are the ease of overtraining and the difficulty of programming.

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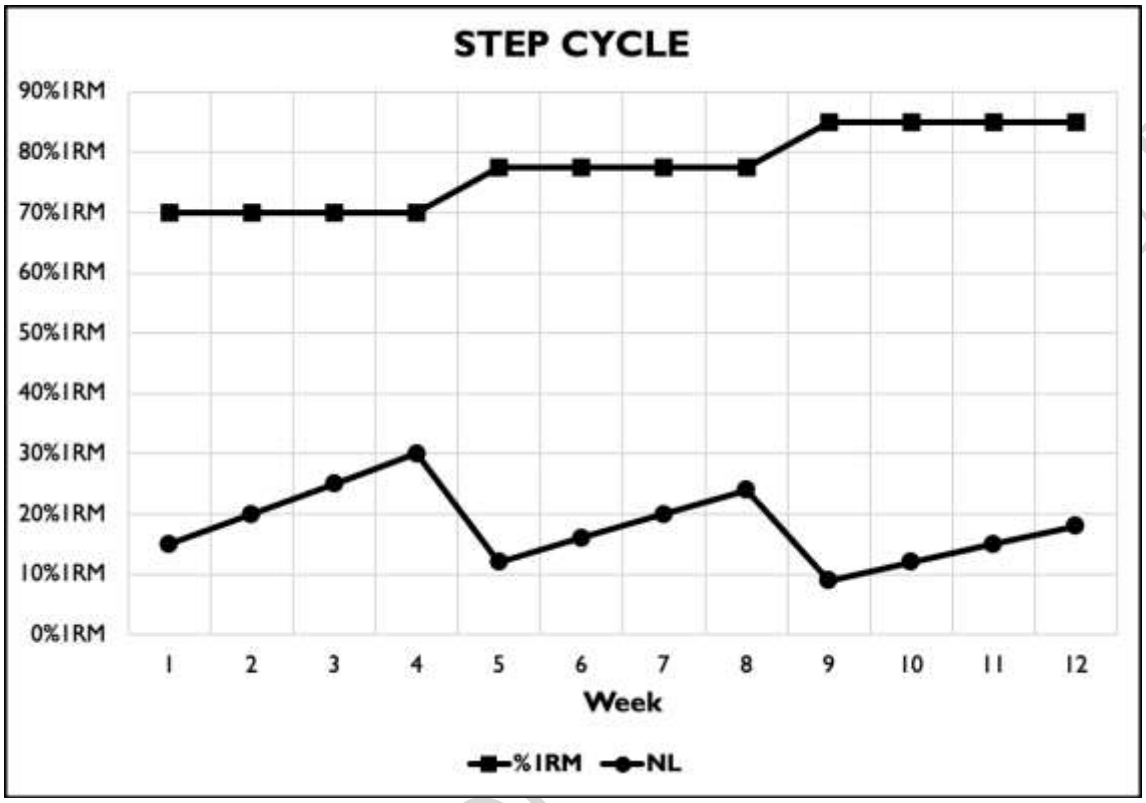
Step cycling

- "The main problem is how to increase the training load…there is the question of whether the loads have to be increased linearly from one training session to another. The alternative possibility is repetition of the same exercise load during a certain time to ensure complete adaptation to this level of exercise load, and only then do a transition to a higher load." (Viru)



- Step loading reverses the RPE dynamics of the linear progression. In the latter, the athlete progresses from ~7 to ~8.5 as the external load increases. In the former the RPE drops from ~8 to ~6 as the internal load decreases.

- Step loading is preferrable for relative beginners, for the beginning of a long training cycle, and for resuming training after injury or illness—in the situations when a very gradual progression is warranted.

  A sharp and simultaneous increase in both volume and intensity is advised as a strong stimulus for high level athletes, although this is no longer traditional progressive overload.

- "The load must be increased whenever it becomes habitual." (Rodionov)

- Typically, the load is stabilized for 1-3 microcycles; 1-week steps are optimal for hypertrophy training.

- Sommer on bodyweight strength training: 8-12 weeks steps, progressing from "hard" through "medium" to "easy" effort.

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- Sharp load increases are preferrable to gradual. The sharper the increase, the longer the stabilization phase.

- Introducing deloading days and microcycles allows for much higher "steps."

  Following are examples of step cycles.

StrongFirst beginners' plan 518			
Week	**Monday**	**Wednesday**	**Friday**
1	"X" x 3/5	"X" x 3/5	"X" x 3/5
2	"X" x 4/5	"X" x 4/5	"X" x 4/5
3	"X" x 5/5	"X" x 5/5	"X" x 5/5
Then add weight and start over:			
1	**"X+Y"** x 3/5	**"X+Y"** x 3/5	**"X+Y"** x 3/5

Boutenko's BP cycle	
Weeks	**Load**
1-4	60% x 10/4
5-8	70% x 6/5
9-12	80% x 5/6
13-16	90% x 3/3

Beyond progressive overload: concentrated loading and delayed training effect

- in *concentrated loading* (Verkhoshansky), fatigue is purposefully accumulated through increased loads and decreased recovery in microcycles and mesocycles. This leads to a temporary performance decrease. It later rebounds following a deload. It is a risky and highly advanced method.

- A classic block of concentrated strength loading consists of 3-4 stress microcycles followed by 2 recovery microcycles. For a load to qualify as "concentrated," the monthly volume must make up 23-25% of the annual volume. (Verkhoshansky)

- A planning tactic used by many top athletes is to divide the 6-8 weeks before an important competition into two particular mesocycles. The first has an exceptionally high volume, intensity, and load variability. Various means are used to further intensify the training process…the use of the competitive method, etc. The second mesocycle features low volume and intensity. Various restoration and active rest means are employed. "As a result, extreme mobilization of the functional resources in the first mesocycle is realized in the adaptive reconstruction in the second mesocycle." (Platonov)

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- Reducing the intensity and emphasizing speed accelerates athletes' recovery.

  E.g., in Smolov's squat plan, a 2-week "switching cycle" of free style explosive work is sandwiched between a concentrated loading cycle and a demanding peaking cycle.

- Although there are successful progressive overload applications of concentrated loading, e.g., "Soviet Squat Routine" and Smolov's squat program, it is best suited for variable loading.

"Soviet squat routine" (Maslayev)		
Week	**Light day**	**Heavy day**
1	70% x 2, 75% x 2, 80% x 2/6	80% x 3/6
2		80% x 4/6
3		80% x 5/6
4		80% x 6/6
5		85% x 5/5
6		90% x 4/4
7		95% x 3/3
8		100% x 2/2
9		105% x 1

The original plan was 6 weeks long, called for squatting 3 times a week, alternating light and heavy days. "I believe it to be the finest strength training system ever brought into this country—bar none." (Hatfield)

Base squat mesocycle (Smolov)				
Week	**Monday**	**Wednesday**	**Friday**	**Saturday**
1	70% 1RM x 9/4	75% x 7/5	80% x 5/7	85%x3/10
2	+10kg			
3	+5kg			
4	Rest		**Test 1RM**	

Smolov's base cycle delivers a 10-30kg gain for heavyweights and 5-7,5kg for lighter lifters. It is followed by a 2-week deloading "switching cycle" that features free style explosive lifting and a peaking cycle.

- When planning concentrated loading cycles, the following guidelines (Dyachkov) are helpful:

  - In a large load week—2 large load days
  - In a medium load week—1 large load day
  - In a light load week—0 large load days; intensity reduced

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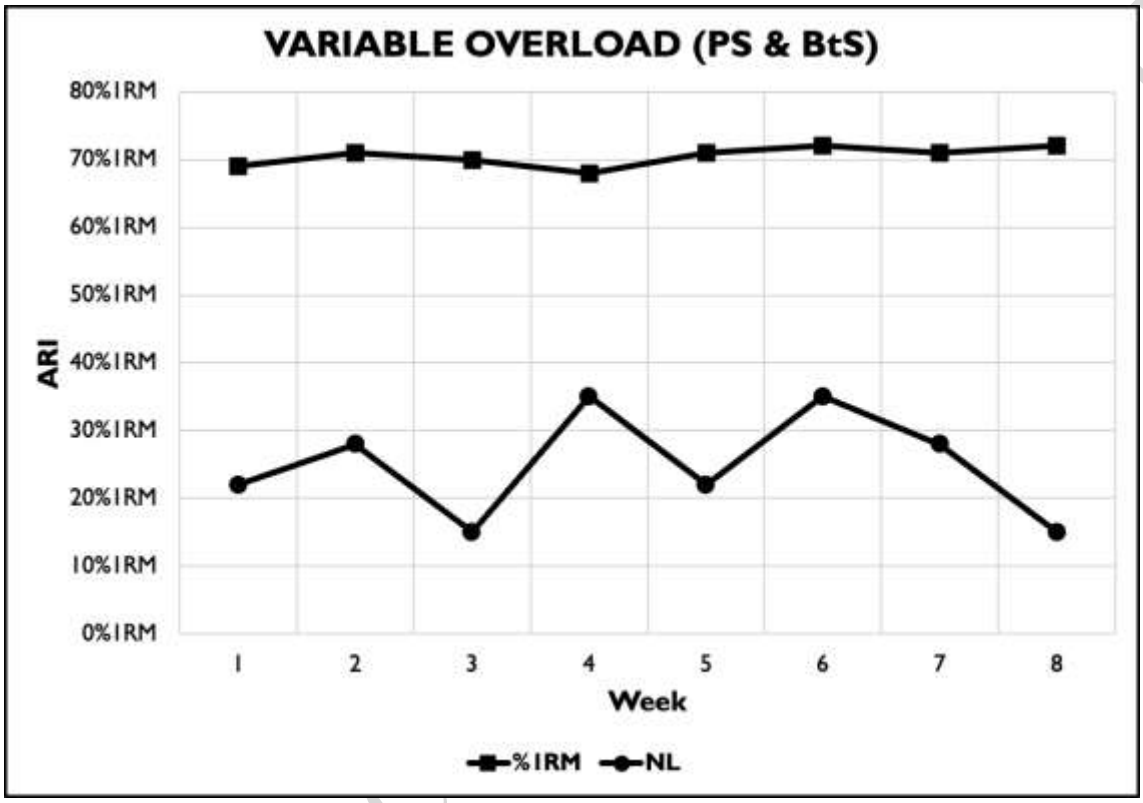
Variable loading

- Ermakov conducted a series of 8-10 week experiments comparing different weekly and monthly load distribution for weightlifters: traditional, smoother dynamics (Matveev, Ozolin) with much sharper changes (Vorobyev). The experimental groups made strength gains 61% greater than the control groups. The researcher also discovered that athletes in the experimental groups were healthier and felt better. Those who used more of a gradual progressive overload sometimes came to training sessions not feeling well and unable to perform the required workload.



- To learn how to program using variable loading, refer to Plan Strong™ and Built Strong seminars.

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Fitting progressive overload mesocycles together

- Keep up the momentum. Frequent alternation of adaptation and deadaptation excessively strains the genetically determined adaptive capacity. Maintenance with moderate loads is incomparably superior.

- The type of the next mesocycle after a progressive overload developmental mesocycle such as the ones described earlier depends on the immediate training goals.

  - If the athlete is not planning on competing or maxing, build another developmental mesocycle. Make a couple of changes while keeping many constants, including the progression rhythm. E.g.:

    - Switch the main lift to an SV exercise
    - Change the exercise dynamics, e.g., add pauses
    - Choose a different variable to progress, e.g., add reps instead of weight
    - Stick with increasing the chosen variable once a week for 6 weeks

  - *Block periodization* or *block training* (BT) is an option.

    Alternate 2-4 week blocks dedicated to hypertrophy and neural adaptations. The BT model takes advantage of *training residuals*, the fact that strength adaptations, both morphological and neurological, are largely retained for ~30 days. Alternating training targets allows to build up one without losing ground in the other. In addition, a time off a particular exercise and/or training load resensitizes the organism to it.

    Since 2-4 weeks are not long enough to form stable adaptations, repeat the same exercise variations and dynamics several blocks in a row. You may change the quantitative load parameters but stay close to the same intensity zone(s).

If the athlete is going to compete or test his PR, there are at least two options.

1. *Classic US powerlifting big cycle*. Design a cycle that strings on several 4-6 week mesocycles back to back, increasing the intensity towards the competition in a linear or wave manner while reducing the volume. The last mesocycle is for peaking; remember not to exceed 90% 1RM for more than 3 weeks, including the meet day. Also remember that after competition or a maximal intensity training session the CNS needs at least three weeks to recover, which necessitates light and easy training in the transition period.

   E.g., a deadlift big cycle by Gallagher:

Week	Reps per set
1-4	6
5-8	4
9-12	2

   Even though the intensity goes up every week, the first week of each 4-week block acts as a deload due to the rep decrease (and added supportive gear for powerlifters).

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2. *Conjugate-sequential system* (Verkhoshansky).

   "Conjugate" refers to developing multiple qualities with the same means (Dyachkov); in this case, strength and skill. "Sequential" refers to the arrangement of progressively more effective training means in time.

   The system was developed for SPP but it its applicable to general strength training. E.g., in a bench press cycle, you could incorporate assisted supramaximal singles with professional spotting and no psyching into the peaking mesocycle. E.g., if you are training for the Iron Maiden/Beast Tamer pullup, you could finish a big cycle with 6 weeks of isometrics.

Microcycle/week

- A *microcycle* is 3-14 days long; a week is most common.

- Types of microcycles:

  - *Introductory*
  - *Loading*
  - *Stress*
  - *Precompetition*
  - *Competition*
  - *Restorative*

- Classification of training sessions by load:

  - *Developmental (heavy)*—enough to improve the targeted quality after supercompensation.
  - *Stabilization (medium)*—often the same intensity as the developmental load but a lower volume.
  - *Restorative (light)*—lower than a developmental load in both volume and intensity.

- Different types of rest intervals between sessions (Matveev):

  - *Ordinary*—recovers the same work capacity.

  - *Stiff*—under recovery, summation of effects of several sessions for a stronger stimulus. "Experience convinces that it is one of the main ways of increasing effectiveness of training sessions enabling one to reach extraordinary results. But it is justified only under strictly defined conditions with the most important being—qualified regulation of loads and recovery."[6] Otherwise overtraining.

  - *Supercompensatory*—long enough to improve the target quality after a developmental session.

The interval between developmental sessions must be long enough to enable supercompensation. How to determine if supercompensation has taken place: the athlete exceeds last developmental session's results at the same level of effort. (Kozhourkin)

[6] Appropriate for concentrated and variable loading; rarely used for progressive loading.

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- A safe frequency for developmental (heavy) loads for a given quality is once every 5-7 days.
- A time tested weekly template for a given lift (modified York Barbell):
  - 3 nonconsecutive training days a week, e.g., Monday, Wednesday, Friday.
  - The heavy day is in the middle of the week, preceded by the light days 2 days earlier. This takes advantage of the facts that one's work capacity peaks in the middle of the week and that a light load 48 hours earlier has a facilitation effect. Thus, Monday—light, Wednesday—heavy, Friday—medium.
  - Make the volume of the light day 60% of that of the heavy day (the "60% rule") and the volume of the medium day 80%.
  - Compared to the heavy day, maintain the intensity on the medium day and reduce on the light day.
  - A fourth day (Saturday) may be added for assistance work.
  - Miscellaneous exercises for small muscle groups may be done on Tuesdays and Thursdays.
- Another effective weekly template (Vorobyev):
  - 4 days a week, with 2 training days back to back, e.g., Monday, Tuesday, Thursday, Friday.

Light days

- Cons:
  - Less time efficient
  - Interference with conversion of type IIA fibers into IIX

  Pros:
  - More skill practice
  - Compliance with the waviness principle and the training continuity principle
  - An anabolic stimulus (Vorobyev)
  - "...reduced training ensures better actualization of previous training effects than complete rest." (Viru)
  - Neural strength gains through synaptic facilitation and myelination
  - Increased CP stores = ability to handle heavy loads for more sets on heavy days. In addition, increasing CP capacity is likely to promote myofibrillar hypertrophy: "The amount of actin in myofibrils correlates linearly with the concentration of creatine in the muscles." (Volkov)
  - Increased glycogen stores
  - Keeps a motivated athlete out of trouble—prevents "random acts of variety"
- Different ways to organize a light day:
  1. **Same exercise, reduced volume and intensity.** The reps per set and the weights are typically lowered by 20-30%. *(E.g., after 300x5/5 on the heavy day do 225x4/5).*

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Towards the end of a cycle, this tactic does not work. *(E.g., after a heavy day of 335x3/3, 245x2/2 would be of little value.)* Maintain the light day loads from the higher volume phase of the cycle. *(E.g., 225x4/5)*

Another option is to maintain the same load in all light days of the cycle. *(E.g., 50-60% x 5/5, 40-50% x 10/3, etc.)* Recording RPE will enable you to track your progress.

2. **Same exercise, significantly reduced intensity but increased volume**—a hypertrophy emphasis. Stop far from failure. *(E.g., 225x10/1-2 or 245x5-6/3-5 after a heavy day with 315x3/3)* This type of a light day can be done sooner than usual—2-3 days after the heavy day.

3. **Assistance work only.** Typically, several sets of 4-6 reps with ~RPE 8.

    a. A variation (SV) of the primary lift.

    b. Another exercise or other exercises for the same muscle groups. *(E.g., dips or military press for the BP)*

4. **"Stealth" light day.** Considering the overlap and the synergy between the SQ and the DL or the BP and the MP, a heavy day for one becomes a light day for the other. This is the "no light days" approach of the 1980s and 1990s US powerlifting cycling methodology.

Relative to other exercises, light days may be organized as follows:

1. **Combine a light day for one lift with a heavy day for another lift.** *E.g., SQ+BP assistance, BP+DL assistance, SQ+DL assistance.* This was a common practice among Soviet powerlifters in the 1980s.

2. **Combine light days for two different lifts.** *E.g., SQ+BP; SQ+DL.*

- A sample cycle with light/medium days:

StrongFirst kettlebell military press and hypertrophy plan 907			
Week	**Monday**	**Wednesday**	**Friday**
1	**(2, 4, 6) x 2**	(2, 4, 6) x 2	**(2, 6) x 3**
2	(2, 6) x 3	**(2, 4, 6) x 3**	(2, 4, 6) x 3
3	**(2, 6) x 4**	(2, 6) x 4	**(2, 4, 6) x 4**
4	(2, 4, 6) x 4	**(2, 6) x 5**	(2, 6) x 5
5	**6/6**	6/6	**RM Test**
Heavy days (bold): 7-8RM kettlebell. Medium days: match the sets and reps of the last heavy day with a lighter kettlebell: -4kg for women and -8kg for men. One clean, multiple presses. On heavy days rest long enough for the heart rate to come down to normal. Take an extra minute before sets of 6. On medium days keep the rest periods short but not challenging.			

- A+A with strength exercises (Strength Aerobics) can serve as a light day for strength training: 70% 1RM x 3/≤15, 1min rest between sets.

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Hypertrophy training

- Soviet weightlifters of all levels were advised to allow the muscle mass to grow organically and were warned against accelerating the process.

- A high training volume disrupts the metabolism and activates trophic processes. (Vasiliev)

  "Everything else being equal, an increase in the training volume facilitates a large increase in muscle mass." (Roman)

  However, above a certain volume threshold, the adaptations favor energy production over protein synthesis in drug free athletes. (Vorobyev)

- To stimulate hypertrophy, increase the share of sets of 5-6 reps. (Vorobyev) 70-75% 1RM intensity was common.

- An additional muscle building tactic used by heavyweight weightlifters is following up the main workload with several sets of 5-10 reps with 50-65% 1RM in squats, bench presses, or military presses. (Medvedev)

- A common weightlifting tactic of increasing the volume towards the end of the week is applicable to hypertrophy training.

- "The best conditions for anabolic processes are when small (2-4 ton) loads with low intensity follow large loads." (Vorobyev)

- It is preferable to schedule muscle hypertrophy sessions in the end of a training session and in the evening.

  Avoid endurance training for 24-36 hours after a hypertrophy training session.

- In the conventional periodization model, the athlete works on hypertrophy in the beginning of the cycle and trains the nervous system closer to the peak.

  Muscles take longer to adapt than the nervous system, thus training cycles focusing on hypertrophy should be longer than those aiming at neural strength adaptations. E.g., two 6-weeks of hypertrophy training followed by one 6-week block of strength emphasis.

- The block periodization model alternates morphology and neurology every mesocycle (4 weeks) or even a half-mesocycle (2 weeks). This model was pioneered by the Soviets in the 1970s. (Ivanova et al.)

  In powerlifting, alternating 4-week blocks of hypertrophy and strength emphasis can be recommended to beginners and to lifters of all levels off season.

  However, the above strategy will work only if one keeps revisiting the same training means and methods; recall that changing exercises more often than once a month leads to transient adaptations.

  Also consider the training residuals: maximal strength, both the neural and the hypertrophy adaptations, is maintained for 30±5 days.

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Easy Strength Classic 2.0

1. Technique

Do not "work out"; "practice." Stay tight.

2. Exercises

2-3 global and regional barbell strength exercises (power cleans and clean pulls also acceptable)

3. Frequency

2-3/week

4. Intensity

80-95% 1RM (~2-8RM)

5. Speed

Medium

6. Effort

Stop a set before the bar starts slowing down. Limit the RPE to 7-8. Finish lifting feeling stronger than when you started.

7. Reps

1-6

8. Rest periods

Ordinary, typically 3-5min

9. Volume

NL 10-20 for "easy strength"; 20-30 for better results

10. Progression and load variability

a) Change the intensity every training session. Cycle or vary more informally and even randomly. If the latter, change the set and rep scheme every time.
b) Practice specialized variety.
c) If you need to deload, reduce the intensity to 70% 1RM and emphasize speed.
d) Go for a PR, single or rep, when you are feeling exceptionally strong but stop short of an all-out max. Set a "sort of max" at RPE 9. Always back off after a PR for at least two weeks.
e) Do not stop strength training in season but reduce the volume by 1/2-2/3. You may switch from 3 to 2 strength sessions a week.

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Plan Strong™ Tactical

Progressive overload methods rely on a controlled, predictable environment, which makes them a poor choice for military and first responders who face unpredictable schedules and high physical and psychological demands. Variable overload is the only logical choice.

Plan Strong™ Tactical applies select elements of the Soviet weightlifting training methodology to highly kinetic environments.

1. **Optimize daily load parameters according to Plan Strong 70 preparatory period guidelines.**
2. **Obey the StrongFirst Stop Signs.**
3. **Train almost daily; when possible, "grease the groove."**
4. **Uncouple volume and intensity.**
5. **Practice sharp load variability, driven by the Delta 20 principle, the current professional demands, randomness within the defined limits, and understanding of summation of different loads.**
6. **Periodically use concentrated loading.**
7. **Set opportunistic PRs.**

Monthly volume distribution in the high intensity zones			
% 1RM	**70-80%**	**81-90%**	**91-100%**
Share of intensity zones in the monthly volume	50-60	30-40	0-10

NL per class of exercise per day, ≥70% 1RM		
Minimal	**Optimal**	**Maximal**
10-20	20-30	30-50

Repetitions per set in different intensity zone		
Warm-up zone <70%	**Training zone 70-90%**	**Control zone >90%**
4-10	1/3-2/3 RM	1
Use ordinary rest periods—rest until you feel fully recovered.		

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Plan Strong™ Algorithm 1118D

The algorithm is for an intermediate lifter training two barbell lifts: a press (military, bench, floor), and either a squat (Zercher, front, back), or a deadlift (conventional, sumo). If you choose to do both squats and deadlifts, alternate them every 6 weeks.

Plan Strong™ algorithm 1118 was designed to deliver optimal training loads with high variability. It is to be used on an ongoing basis. It is not meant to be used to peak on a given date. The algorithm is compatible with most athletic training.

Load variability is achieved by rolling a die, separately for each lift.

Step 1: Training days

Train the two lifts on the same days (in the same session or in separate sessions), on different days, or a combination of the above. The choice of the training days is up to you; you may vary them every week.

Number of training days for the given week: Press						
Die shows	**1**	**2**	**3**	**4**	**5**	**6**
Training days	2	3				4

Number of training days for the given week: Deadlift						
Die shows	**1**	**2**	**3**	**4**	**5**	**6**
Training days	1	2				3

Number of training days for the given week: Squat						
Die shows	**1**	**2**	**3**	**4**	**5**	**6**
Training days	1	2		3		

If you are training for a sport or are engaged in demanding endurance training, try to abide by the following guidelines:

- Plan ≥90% 1RM sessions when you are fresh.
- Avoid endurance training for 24 hours after higher volume (NL ≥ 20) loads.

Obviously, it will not always be possible; do your best. Note that, even though your performance on heavy days performed in a fatigued state will suffer, you are still getting stronger.

On the days when you are training two lifts, start with the lift for which heavier weights were prescribed. If the intensity is the same for both lifts, vary the order.

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E.g., you have a martial art practice on Monday and Wednesday nights and Saturday mornings. The die prescribed the following loads, in this order for the upcoming week: ***BP—1) 70% 1RM (NL 25); 2) 80% 1RM (NL 15); 3) >90% 1RM; DL—1) 70% 1RM (NL 30); 2) 85% 1RM (NL 7)****. Start by placing the >90% 1RM BP session, the last of the week. Then assign the rest of the loads. Some of the many options are listed below.*

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM	DL 70% (30) BP 70% (25)		BP 80% (15)			MA	Rest
PM	MA		MA		BP >90% DL 85% (7)		

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM	BP 70% (25)		BP 80% (15)			MA	Rest
PM	MA	DL 70% (30)	MA		BP >90% DL 85% (7)		

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM		BP 70% (25) DL 70% (30)	BP 80% (15)			MA	Rest
PM	MA		MA		BP >90% DL 85% (7)		

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM	BP 70% (25) DL 70% (30)					MA	Rest
PM	MA	BP 80% (15)	MA	DL 85% (7)	BP >90%		

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM	BP 70% (25) DL 70% (30)		BP 80% (15)			MA	Rest
PM	MA		MA	DL 85% (7)	BP >90%		

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM	DL 70% (30)	BP 70% (25)	BP 80% (15)		DL 85% (7)	MA	Rest
PM	MA		MA		BP >90%		

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Step 2: Intensity

Intensity						
Die shows	**1**	**2**	**3**	**4**	**5**	**6**
Weight, % 1RM	70	75		80		≥85
If the roll of a die prescribes an intensity identical to the one in the last session, roll again. If you lifted ≥90% within the last 7 days and you rolled a six (≥85%), roll again according to the above algorithm (your choices are limited to 70, 75, and 80%). Otherwise, if you rolled a six (≥85%), roll again according to the algorithm below:						
Die shows	**1**	**2**	**3**	**4**	**5**	**6**
Weight, % 1RM	85		90		>90	
If after your top warm-up sets you realize that you are not going to make the day's plan, reduce the prescribed intensity by 5-10% while sticking to the prescribed volume.						

Step 3: Volume

NL for different intensities						
Die shows	**1**	**2**	**3**	**4**	**5**	**6**
70% 1RM	20	25			30	
75% 1RM	15	20			25	
80% 1RM	10	15			20	
85% 1RM	5		7			10
90% 1RM	3				5	
If a roll of a die prescribes the NL identical to the one in the last training session (regardless of the intensity), roll again.						
>90% 1RM	Do not roll a die. Work up to a comfortably heavy single. If you are feeling exceptionally strong, try a PR. If you set a PR, recalculate your 70-90% 1RM based on it.					

Step 4: Speed

Concentric speed with 70-85% 1RM						
Die shows	**1**	**2**	**3**	**4**	**5**	**6**
Bar speed	Fast		Medium			Slow
"Fast"—as fast as possible without jerking. Use compensatory acceleration. "Medium"—a comfortable, confident speed, corresponding to a first attempt in competition. "Slow"—corresponds to the speed of a hard-fought PR single. With >85% weights use a medium speed. In all cases use a medium eccentric speed.						

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Warm-up

Perform a non-fatiguing warm-up of choice. Use the following rep ranges in your warm-up sets:

Repetitions per set in warm-up sets										
% 1RM	**50%**	**55%**	**60%**	**65%**	**70%**	**75%**	**80%**	**85%**	**90%**	**95%**
Repetitions per set	3-5		1-3			1-2		1		

Reps per set

There is no roll of a die to determine the number of reps per set.

Repetitions per set in work sets					
% 1RM	**70%**	**75%**	**80%**	**85%**	**≥90%**
Repetitions per set	3-6	3-5	2-4	2-3	1

Vary the reps within the specified ranges as much as possible.

> *E.g., today's load is 70% 1RM, NL 20. Some of the many possibilities in the 3-6 rep range: (4, 6, 4, 6); (3, 5, 3, 5, 4); (3, 6, 3, 5, 3); (3, 4, 5, 3, 5), etc.*

Rest periods between sets

Rest until you feel totally ready to lift—then wait one more minute—with weights up to 90% 1RM. Do not compress the rest periods; this is not a "metcon"!

Rest for 5min or longer with >90% weights.

Power to you!

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Easy Strength through Easy Power

(Based on Hatfield)

Die shows	**1**	**2**	**3**	**4**	**5**	**6**
% 1RM	65		70			75
If you rolled the same number as the last session, roll again. If your other training or circumstances demand it, do not roll a die, and lift 60% 1RM.						
Do 5/5 with compensatory acceleration.						
Rest for 5min between sets.						
When 75% starts moving noticeably faster, retest your 1RM the next time.						
Train each lift from 3 times in 2 weeks to 2 times a week.						
Approximately once a month after the 5/5 rest for 5min, then do ~5 singles every 1-2min with ~80-90% 1RM.						

Instead of the above variable loading algorithm you may practice linear, wave, or step cycling. E.g., keep repeating 65-70-75% until 75% moves faster. Follow up with a short peaking cycle.

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The Easy Strength Package

This program is aimed at athletes experienced with the barbell but not competitive lifters.

- Select two main lifts. Progress them independently.
- Lift at a medium speed, regardless of the exercise and the weight.
- Train each lift 3 times a week; the deadlift 2 times.
- Rest between sets for 5min unless specified otherwise.
- Rotate these three cycles:

    1. **Power to the People! 3.0**
    2. **Stairway to Strength (S2S)**
    3. **Russian Countdown**

- Insert a week of "random variety" between the above three cycles.
- When you feel deep fatigue building up, pause the cycles and insert 6 weeks of "Rest & Reenforce" training.

Power to the People! 3.0

Starting weight

10-12RM (an estimate is acceptable).

Progression

Stay with the same weight for three heavy days before adding more: a week and a half for deadlifts, a week for other lifts; longer if some heavy days were replaced with light days. The amount to add:

Weight Increases, "X"			
Kgs		Lbs	
1RM	Add	1RM	Add
50-150kg	5kg	100-330lb	10lb
150-200kg	7.5kg	330-440lb	15lb
>200kg	10kg	>440lb	20lb

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Phase I: 6/3 (NBL 18)

Add "X" amount of weight every three heavy days until you reach RPE 8 on your last set. Stabilize at that intensity for three heavy days.

Phase II: 6, 4, 2 (NBL 12)

Keep adding "X" amount of weight every three heavy days until you reach RPE 8 on your first set. Stabilize at that weight for three heavy days. You might will reach RPE 8 on your first day with this set and rep scheme. Stabilize at that weight.

Phase III: 4, 2 → 2 (NBL ≤ 6)

- There are two heavy days in this phase:

  1) Add "X" and do 4, 2 reps.
  2) Add "X" and do 2 reps.

1RM testing

- Test your 1RM 4 days after the last day of Phase III.

  2 days before the test, warm up as follows: 50% 1RM x 5/2, 60 x 3/2, 70 x 2/2. If your experience tells you to take a different number of days off and a different taper, use your proven formula.

Light days

Every other training day is light. In addition, take an unplanned light day if your heaviest ramp-up set or the first work set feels unusually heavy.

Introduce light days on session #5, after the first weight increase.

The light day's dynamics follow the "pendulum principle": its weight goes down by the same amount ("X") as the heavy day's goes up.

Then the set number stays at three while a rep is added every time the weight drops.

E.g., 100kg x 6/3, 105kg x 7/3, 110kg x 8/3, etc.

The following sample table will make it clear.

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A sample progression for an athlete with a back squat of 405x1RM, 305x10RM, and X=15lb		
Week #	***Session #***	***Load***
I	*1.*	***305x6/3 (~75% 1RM)***
	2.	***305x6/3 (~75% 1RM)***
	3.	***305x6/3 (~75% 1RM)***
II	*4.*	***320x6/3 (~79% 1RM)***
	5.	*290x6/3 (~72% 1RM) (light day)*
	6.	***320x6/3 (~79% 1RM)***
III	*7.*	*290x6/3 (~72% 1RM)*
	8.	***320x6/3 (~79% 1RM)***
	9.	*290x6/3 (~72% 1RM)*
IV	*10.*	***335x6/3 (~83% 1RM), RPE 8***
	11.	*275x7/3 (~68% 1RM)*
	12.	*275x6/3 (~68% 1RM)—an unplanned light day; the warm-ups felt heavy*
V	*13.*	***335x6/3 (~83% 1RM)***
	14.	*275x7/3 (~68% 1RM)*
	15.	***335x6/3 (~83% 1RM)***
VI	*16.*	*275x7/3 (~68% 1RM)*
	17.	***350x6, 4, 2 (~86% 1RM), RPE 8***
	18.	*260x8/3 (~64% 1RM)*
VII	*19.*	***350x6, 4, 2 (~86% 1RM)***
	20.	*260x8/3 (~64% 1RM)*
	21.	***350x6, 4, 2 (~86% 1RM)***
VIII	*22.*	*260x8/3 (~64% 1RM)*
	23.	***365x4, 2 (~90% 1RM)***
	24.	*245x9/3 (~60% 1RM)*
IX	*25.*	***380x4 (~94% 1RM)***
	26.	*230x10/3 (~57% 1RM)*
	27.	***395x2 (~98% 1RM)***
X	*28.*	*205x5/2, 245x3/2, 285x2/—taper*
	29.	***420x1RM (~104% 1RM, PR)—test***

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Stairway to Strength (S2S)

Starting weight

The weight you have completed Phase II of PTP 3.0 with.

To stay with the above example, 350lb.

Progression

The reps and the sets stay at (1, 2, 3) x 3 (NBL 18), except for the last set once every several weeks as explained below. Rest for 5min before each triple, 2-3min before singles and doubles.

Every third heavy day—once in 3 weeks for deadlifts, 2 weeks for other lifts; longer if some heavy days were replaced with light days—take the last set to RM. If you have made ≥5 reps, next time increase the weight by 5-10lb (~1/2 X). If you have not, the cycle is over.

Light days

Every other training day is light. In addition, take an unplanned light day if your heaviest ramp-up set or the first work set feels unusually heavy.

Perform 6/3 with the weight you started the last PTP 3.0 cycle with. In this plan the light day's load parameters do not change throughout the cycle.

To stay with the above example, 305x6/3.

The entire progression might look like this:

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Week #	Session #	Load
I	1.	**(350lb x 1, 2, 3) x 3**
	2.	305x6/3
	3.	**(350lb x 1, 2, 3) x 3**
II	4.	305x6/3
	5.	**350lb x 1, 2, 3, 1, 2, 3, 1, 2, 7RM**
	6.	305x6/3
III	7.	**(360lb x 1, 2, 3) x 3**
	8.	305x6/3
	9.	**(360lb x 1, 2, 3) x 3**
IV	10.	305x6/3
	11.	**360lb x 1, 2, 3, 1, 2, 3, 1, 2, 5RM**
	12.	305x6/3
V	13.	**(370lb x 1, 2, 3) x 3**
	14.	305x6/3
	15.	**(370lb x 1, 2, 3) x 3**
VI	16.	305x6/3
	17.	**370lb x 1, 2, 3, 1, 2, 3, 1, 2, 4RM** —the cycle is over.

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The Russian Countdown

Starting weight

The last RM weight from the S2S cycle in which you managed ≥5 reps.

To stay with our example, 360lb.

Progression

The progression is straight out of the Fighter Pullup Program:

Session #	Reps	NBL
1.	4, 3, 2, 1, 1	11
2.	4, 3, 2, **2**, 1	12
3.	4, 3, **3**, 2, 1	13
4.	4, **4**, 3, 2, 1	14
5.	**5**, 4, 3, 2, 1	15
6.	5, 4, 3, 2, **2**	16
7.	5, 4, 3, **3**, 2	17
8.	5, 4, **4**, 3, 2	18
9.	5, **5**, 4, 3, 2	19
10.	**6**, 5, 4, 3, 2	20

Once you have reached 6, 5, 4, 3, 2 reps, add 5-10lb (~1/2 X) and start over with 4, 3, 2, 1, 1.

The plan is over when you cannot make the required sets with perfect form and RPE 8.

Light days

If your heaviest ramp-up set or the first work set feels unusually heavy, lift 70% 1RM x 6/3 with 3min rests.

Testing

Rest for a day, have a light day, rest for another day, and test your 10-12RM to set the starting weight for the next PTP 3.0 cycle.

Example:

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Week #	Session #	Load
I	1.	360x4, 3, 2, 1, 1
	2.	360x4, 3, 2, **2**, 1
	3.	360x4, 3, **3**, 2, 1
II	4.	360x4, **4**, 3, 2, 1
	5.	360x**5**, 4, 3, 2, 1
	6.	360x5, 4, 3, 2, **2**
III	7.	360x5, 4, 3, **3**, 2
	8.	360x5, 4, **4**, 3, 2
	9.	360x5, **5**, 4, 3, 2
IV	10.	360x**6**, 5, 4, 3, 2
	11.	370x4, 3, 2, 1, 1
	12.	370x4, 3, 2, **2**, 1
V	13.	370x4, 3, **3**, 2, 1
	14.	370x4, **4**, 3, 2, 1
	15.	370x**5**, 4, 3, 2, 1
VI	16.	The first set felt hard. Stopped after 2 reps and had a light day: 295x6/3.
	17.	370x5, 4, 3, 2, **2**
	18.	370x5, 4, 3, **3**, 2
VII	19.	370x5, 4, **4**, 3, 2
	20.	370x5, **5**, 4, 3, 2
	21.	Did not feel strong during the warm-up and decided to have a light day: 295x6/3.
VIII	22.	370x**6**, 5, 4, 3, 2—The session was hard; time to stop the cycle.
	23.	295x6/3—the light day before the testing.
	24.	**Test: 335x10RM**—Start your next PTP 3.0 cycle with this weight.

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Rest & Reenforce (R&R)

When you feel deep fatigue building up, insert 6 weeks of "Rest & Reenforce" training for both lifts.

- Train 3 times a week. Limit each session to 45min, excluding the warm-up and the cool-down.
- Pick 0-2 exercises per muscle group and stick with them for 6 weeks.

  Your options are SV exercises for your main lift (if you are not mentally fatigued from the last cycle) and "bodybuilding" exercises, multi-joint and single-joint.

  Try to pick exercises that address your weaknesses but do not overthink it.
- "Work up" in sets of 5 or 6 reps until reaching a weight that is comfortably heavy (~RPE 7 or 3 RIR), then perform two more sets of 5-6 reps with this weight. Stay with the same weight for a week.
- Rest between sets: until your breathing has totally normalized plus 1min. When possible, alternate sets of two exercises.
- After your "R&R" resume your previous cycles, starting two weeks back from where you left off.

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Block Blast

(Zonin)

Based on estimated RM at different % 1RM, Block Blast is a 15-week program divided into 4 blocks of 3 weeks each plus a peaking phase, designed to train a single main lift from three different angles:

1. Maximal strength—2–4/5 (reps/sets)
2. Myofibrillar hypertrophy—5/5
3. Sarcoplasmic hypertrophy or power—same training loads but different execution:
    a. Sarcoplasmic hypertrophy: 6–10/5 (reps/sets).
    b. Power: VBT/4–6—maximal concentric speed, end set when speed drops.

Program flexibility

You can run the program with two weekly sessions using different combinations of foci:

1. Maximal strength + myofibrillar hypertrophy
2. Maximal strength + power
3. Myofibrillar + sarcoplasmic hypertrophy

Load selection

The test of repetitions at 80% 1RM (RM@80%) defines the personalized progression. The following tables link each RM@80% result to weekly loads for each session type:

Weekly load progression: Maximal Strength

Week	RM@80%							
	5	6	7	8	9	10	11	12
1–3	60.0%	68.0%	73.3%	77.1%	80.0%	82.2%	84.0%	85.5%
4–6	65.0%	72.0%	76.7%	80.0%	82.5%	84.4%	86.0%	87.3%
7–9	70.0%	76.0%	80.0%	82.9%	85.0%	86.7%	88.0%	89.1%
10–12	75.0%	80.0%	83.3%	85.7%	87.5%	88.9%	90.0%	90.9%
13	80.0%	84.0%	86.7%	88.6%	90.0%	91.1%	92.0%	92.7%
14	85.0%	88.0%	90.0%	91.4%	92.5%	93.3%	94.0%	94.5%
15	TEST!							

Maximal strength progression:

- Fixed 5 sets
- Reps follow a 2-3-4 wave pattern
- Repeats every three weeks, but load increases progressively

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Weekly load progression: Myofibrillar Hypertrophy

Week	RM@80%							
	5	6	7	8	9	10	11	12
1–3	45.0%	56.0%	63.3%	68.6%	72.5%	75.6%	78.0%	80.0%
4–6	50.0%	60.0%	66.7%	71.4%	75.0%	77.8%	80.0%	81.8%
7–9	55.0%	64.0%	70.0%	74.3%	77.5%	80.0%	82.0%	83.6%
10–14	60.0%	68.0%	73.3%	77.1%	80.0%	82.2%	84.0%	85.5%

Weekly load progression: Sarcoplasmic Hypertrophy or Power

Week	RM@80%							
	5	6	7	8	9	10	11	12
1–3	30,0%	44,0%	53,3%	60,0%	65,0%	68,9%	72,0%	74,5%
4–6	35,0%	48,0%	56,7%	62,9%	67,5%	71,1%	74,0%	76,4%
7–9	40,0%	52,0%	60,0%	65,7%	70,0%	73,3%	76,0%	78,2%
10–14	45,0%	56,0%	63,3%	68,6%	72,5%	75,6%	78,0%	80,0%

Example:

An athlete with a 1RM back squat of 300lb tests his RM@80% 1RM: 300 × 0.80 = 240lb. He performs 7 perfect reps with this weight. Thus, his RM@80% 1RM = 7, and he selects the corresponding columns in the tables to determine his training intensities across all blocks:

Maximal Strength

Week	%1RM	Calculation	Rounded Load (lb)
1-3	73.3%	300 × 0.733 = 219.9	220
4-6	76.7%	300 × 0.767 = 230.1	230
7-9	80.0%	300 × 0.800 = 240.0	240
10-12	83.3%	300 × 0.833 = 249.9	250
13	86.7%	300 × 0.867 = 260.1	260
14	90.0%	300 × 0.867 = 270.0	270

Myofibrillar Hypertrophy

Week	%1RM	Calculation	Rounded Load (lb)
1-3	63.3%	300 × 0.633 = 189.9	190
4-6	66.7%	300 × 0.667 = 200.1	200
7-9	70.0%	300 × 0.700 = 210.0	210
10-15	73.3%	300 × 0.733 = 219.9	220

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Sarcoplasmic Hypertrophy or Power

Week	%1RM	Calculation	Rounded Load (lb)
1-3	53.3%	300 × 0.533 = 159.9	160
4-6	56.7%	300 × 0.567 = 170.1	170
7-9	60.0%	300 × 0.600 = 180.0	180
10-15	63.3%	300 × 0.633 = 189.9	190

Maximal Strength Load Progression

Week	Load (lb)	Reps/Sets
1	220	2/5
2	220	3/5
3	220	4/5
4	230	2/5
5	230	3/5
6	230	4/5
7	240	2/5
8	240	3/5
9	240	4/5
10	250	2/5
11	250	3/5
12	250	4/5
13	260	3/5
14	270	2/5
15	**New 1RM Test**	-

Myofibrillar Hypertrophy Load Progression

Week	Load (lb)	Reps/Sets
1–3	190	5/5
4–6	200	5/5
7–9	210	5/5
10–15	220	5/5

Sarcoplasmic Hypertrophy Load Progression

Week	Load (lb)	Reps/Sets
1–3	160	6–10/5
4–6	170	6–10/5
7–9	180	6–10/5
10–15	190	6–10/5

Power (VBT) Load Progression

Week	Load (lb)	Reps/Sets
1–3	160	VBT/4-6
4–6	170	VBT/4-6
7–9	180	VBT/4-6
10–15	190	VBT/4-6

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Progressive Swap

(Zonin)

Progressive Swap is a wave cycle lasting 15 weeks. At first glance, it might resemble a step cycle, since the working weights remain constant for blocks of 4 weeks. However, it is more accurately classified as a wave cycle due to the weekly redistribution between heavy and light sets.

Within each 4-week block, the Average Relative Intensity (ARI) increases progressively as more heavy sets replace light ones. When the next block begins, the load increases, but the ARI drops because the volume again shifts toward lighter sets. The ARI then begins to rise again over the following weeks.

This pattern of rising and falling ARI produces a true wave-like progression, which supports steady strength gains while allowing for technical refinement and fatigue management.

Load calculation: Reload 2.0 method

Start by testing the 1RM for the target lift. Perform maximum reps at 80% of 1RM. Based on the number of reps completed, determine the weekly increment (X) using the Reload 2.0 chart:

RM @ 80%	Jump "X" (%1RM)
4	8,9%
5	6,7%
6	5,3%
7	4,4%
8	3,8%
9	3,3%
10	3,0%
11	2,7%
12	2,4%
13	2,2%
14	2,1%
15	1,9%
16	1,8%
17	1,7%
18	1,6%
19	1,5%
20	1,4%

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The X value is calculated as a percentage, then converted into load units (e.g. pounds or kilos). Important: do not round X immediately. Use the exact percentage to calculate the working weights for each week. Only after calculating, round the result to the nearest usable increment (e.g., nearest 5lb or 2.5kg). This increases precision across the whole cycle.

Weekly set progression

Week	Load H	Load L	Total Sets
1	1RM – 5X	1RM – 8X	6L + 6L + 6L + 6L + 6L + 6L
2	1RM – 5X	1RM – 8X	3H + 3H + 6L + 6L + 6L + 6L
3	1RM – 5X	1RM – 8X	3H + 3H + 3H + 3H + 6L + 6L
4	1RM – 5X	1RM – 8X	3H + 3H + 3H + 3H + 3H + 3H
5	1RM – 4X	1RM – 7X	6L + 6L + 6L + 6L + 6L + 6L
6	1RM – 4X	1RM – 7X	3H + 3H + 6L + 6L + 6L + 6L
7	1RM – 4X	1RM – 7X	3H + 3H + 3H + 3H + 6L + 6L
8	1RM – 4X	1RM – 7X	3H + 3H + 3H + 3H + 3H + 3H
9	1RM – 3X	1RM – 6X	6L + 6L + 6L + 6L + 6L + 6L
10	1RM – 3X	1RM – 6X	3H + 3H + 6L + 6L + 6L + 6L
11	1RM – 3X	1RM – 6X	3H + 3H + 3H + 3H + 6L + 6L
12	1RM – 3X	1RM – 6X	3H + 3H + 3H + 3H + 3H + 3H
13	1RM – 2X	1RM – 5X	2H + 2H + 4L + 4L
14	1RM – X	1RM – 4X	1H + 1H + 2L + 2L
15	**New 1RM test**		

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Example:

An athlete with a 1RM bench press of 225lb tests his RM@80% 1RM.

225 × 0.80 = 180lb. He performs 7 perfect reps with this weight. According to the Reload 2.0 chart, RM@80% = 7 corresponds to a jump (X) of 4.4%.

225 × 0.044 = 9.9lb

He uses this value to calculate the training loads, which are then rounded to the nearest 5lb.

Week	**Load H (lb)**	**Load L (lb)**	**Total Sets**
1	225 – (5 × 9.9) = 175.5 → 175	225 – (8 × 9.9) = 145.8 → 145	6L + 6L + 6L + 6L + 6L + 6L
2	175	145	3H + 3H + 6L + 6L + 6L + 6L
3	175	145	3H + 3H + 3H + 3H + 6L + 6L
4	175	145	3H + 3H + 3H + 3H + 3H + 3H
5	225 – (4 × 9.9) = 185.4 → 185	225 – (7 × 9.9) = 156.7 → 155	6L + 6L + 6L + 6L + 6L + 6L
6	185	155	3H + 3H + 6L + 6L + 6L + 6L
7	185	155	3H + 3H + 3H + 3H + 6L + 6L
8	185	155	3H + 3H + 3H + 3H + 3H + 3H
9	225 – (3 × 9.9) = 195.3 → 195	225 – (6 × 9.9) = 165.6 → 165	6L + 6L + 6L + 6L + 6L + 6L
10	195	165	3H + 3H + 6L + 6L + 6L + 6L
11	195	165	3H + 3H + 3H + 3H + 6L + 6L
12	195	165	3H + 3H + 3H + 3H + 3H + 3H
13	225 – (2 × 9.9) = 205.2 → 205	225 – (5 × 9.9) = 175.5 → 175	2H + 2H + 4L + 4L
14	225 – (1 × 9.9) = 215.1 → 215	225 – (4 × 9.9) = 185.4 → 185	1H + 1H + 2L + 2L
15	**New 1RM test**		

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Triple Tier 2.0

(Zonin)

The Triple Tier 2.0 program spans 12 weeks, divided into 3 blocks of 4 weeks each. Each block follows a variable overload pattern based on the Δ20 principle: weekly volume increases from week 1 to week 3 with over 20% change, followed by a deload in week 4.

Average Relative Intensity (ARI) is stable at 71%, thanks to a fixed distribution of weekly training volume across 65%, 75%, and 85% 1RM.

The key variable that changes block by block is the RIR (Repetitions in Reserve), and as a result, the ARE (Average Relative Effort):

- Block 1 uses ~1/3 of the RM
- Block 2 uses ~1/2 of the RM
- Block 3 uses ~2/3 of the RM

As reps per set increase while weekly volume remains constant, the number of sets per session decreases. This leads to shorter sessions and a gradual shift in training focus from strength to hypertrophy.

Weekly structure and volume distribution

Each week includes three training sessions:

- Heavy: 85% 1RM (4–7 RM) → 15% of weekly volume
- Medium: 75% 1RM (8–11 RM) → 30% of weekly volume
- Light: 65% 1RM (12–15 RM) → 55% of weekly volume

Weekly percentages of monthly volume follow the Δ20 principle:

- Week 1: 22%
- Week 2: 28%
- Week 3: 35%
- Week 4: 15% (deload)

RM fraction used per block

Weekly reps per set are based on progressive RM fractions:

- Weeks 1–4 → 1/3 RM
- Weeks 5–8 → 1/2 RM
- Weeks 9–12 → 2/3 RM

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RM fractions and volume distribution

HEAVY DAY				
Week	**% of monthly NL**	**%1RM / RM**	**RM Fraction**	**% of weekly NL**
1	22%	85% (4–7 RM)	1/3	15%
2	28%			
3	35%			
4	15%			
5	22%		1/2	
6	28%			
7	35%			
8	15%			
9	22%		2/3	
10	28%			
11	35%			
12	15%			

MEDIUM DAY				
Week	**% of monthly NL**	**%1RM / RM**	**RM Fraction**	**% of weekly NL**
1	22%	75% (8–11 RM)	1/3	30%
2	28%			
3	35%			
4	15%			
5	22%		1/2	
6	28%			
7	35%			
8	15%			
9	22%		2/3	
10	28%			
11	35%			
12	15%			

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LIGHT DAY				
Week	**% of monthly NL**	**%1RM / RM**	**RM Fraction**	**% of weekly NL**
1	22%	65% (12–15 RM)	1/3	55%
2	28%			
3	35%			
4	15%			
5	22%		1/2	
6	28%			
7	35%			
8	15%			
9	22%		2/3	
10	28%			
11	35%			
12	15%			

Example:

An athlete with a 1RM back squat of 300lb estimates his rep maxes:

- *85% = 5RM*
- *75% = 10RM*
- *65% = 15RM*

He sets a monthly training volume (NL) of 300 reps, distributed across the 4 weeks as:

- *Week 1: 66 reps*
- *Week 2: 84 reps*
- *Week 3: 105 reps*
- *Week 4: 45 reps*

The reps are split across intensity zones as follows:

- *15% Heavy*
- *30% Medium*
- *55% Light*

Reps per set are adjusted by block:

- *85%: 2 / 3 / 3*
- *75%: 3 / 5 / 7*
- *65%: 5 / 8 / 10*

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Week	NL	Heavy Day			Medium Day			Light Day		
		Reps per Set	NL	Reps/ Sets	Reps per Set	NL	Reps/ Sets	Reps per Set	NL	Reps/ Sets
1	66	2	10	2/5	3	20	3/6 + 2	5	36	5/7 + 1
2	84	2	13	2/6 + 1	3	25	3/8 + 1	5	46	5/9 + 1
3	105	2	16	2/8	3	32	3/10 + 2	5	58	5/11 + 3
4	45	2	7	2/3 + 1	3	14	3/4 + 2	5	25	5/5
5	66	2	10	2/5	5	20	5/4	8	36	8/4 + 4
6	84	2	13	2/6 + 1	5	25	5/5	8	46	8/5 + 6
7	105	2	16	2/8	5	32	5/6 + 2	8	58	8/7 + 2
8	45	2	7	2/3 + 1	5	14	5/2 + 4	8	25	8/3 + 1
9	66	3	10	3/3 + 1	7	20	7/2 + 6	10	36	10/3 + 6
10	84	3	13	3/4 + 1	7	25	7/3 + 4	10	46	10/4 + 6
11	105	3	16	3/5 + 1	7	32	7/4 + 4	10	58	10/5 + 8
12	45	3	7	3/2 + 1	7	14	7/2	10	25	10/2 + 5

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Kettlebell Strength Plan 414A

(Kettlebell military press, [weighted] pullup, weighted pistol, and other lifts)

The progression is borrowed from the "Fighter Pullup Program" by an unknown Russian author. The original program is a linear progression. You start with an all-out set—pick the starting point based on your RM—and then cut a rep in each consecutive set for a total of five sets. The next day add a rep to the last set. Then a rep to the set before that, etc.:

Session #	Reps by set	NL
1.	4, 3, 2, 1, 1	11
2.	4, 3, 2, 2, 1	12
3.	4, 3, 3, 2, 1	13
4.	4, 4, 3, 2, 1	14
5.	5, 4, 3, 2, 1	15
6.	5, 4, 3, 2, 2	16
7.	5, 4, 3, 3, 2	17
8.	5, 4, 4, 3, 2	18
9.	5, 5, 4, 3, 2	19
10.	6, 5, 4, 3, 2	20
11.	6, 5, 4, 3, 3	21
12.	6, 5, 4, 4, 3	22
13.	6, 5, 5, 4, 3	23
14.	6, 6, 5, 4, 3	24
15.	7, 6, 5, 4, 3	25
16.	7, 6, 5, 4, 4	26
17.	7, 6, 5, 5, 4	27
18.	7, 6, 6, 5, 4	28
19.	7, 7, 6, 5, 4	29
20.	8, 7, 6, 5, 4	30
21.	8, 7, 6, 5, 5	31
22.	8, 7, 6, 6, 5	32
23.	8, 7, 7, 6, 5	33
24.	8, 8, 7, 6, 5	34
25.	9, 8, 7, 6, 5	35
26.	9, 8, 7, 6, 6	36
27.	9, 8, 7, 7, 6	37
28.	9, 8, 8, 7, 6	38
29.	9, 9, 8, 7, 6	39

The original progression prescribes a day off after five days of training. The plan was intended to be used for four weeks and has an impressive record of increasing one's pullup reps 2.5-3 times after a 2–3 day rest. While viable for relative beginners, this approach would quickly overtrain intermediate and especially advanced strength athletes. Hence the following changes introducing two elements of a powerlifting cycle: starting below your RM and periodically backing off.

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The new plan calls for a 5-8RM starting weight. If you are applying plan 414 to the kettlebell military press, your 1RM must be no lower than 20kg for women and 36kg for men.

Do one clean before multiple presses. The plan should be applied to single presses only; the stress on the back is likely to be too high when repeatedly working close to your rep max.

Start with the rep count one below your RM. For instance, if you can press a given kettlebell for 6RM, start with a set of five. It is session #5 in the table: 5, 4, 3, 2, 1. If you are in the Iron Maiden or Beast Tamer pressing territory, gain more momentum by starting two reps below your RM, e.g., session #1 (4, 3, 2, 1, 1) for 6RM.

Although when cycling, one usually starts a lot lighter than a rep or two below RM, keep in mind that this particular program is meant to push you harder for quick gains for a short, concentrated block of 3-4 weeks.

Press 2-3 days in a row, then take a day off. When the going gets tough or you start feeling fatigue building up, back off to the reps specified 2-3 sessions ago. Your progression might look like this:

Days	***Load***
1.	*5, 4, 3, 2, 1*
2.	*5, 4, 3, 2,* ***2***
3.	*5, 4, 3,* ***3****, 2*
4.	
5.	*5, 4,* ***4****, 3, 2*
6.	*5,* ***5****, 4, 3, 2*
7.	
8.	***6****, 5, 4, 3, 2*
9.	*6, 5, 4, 3,* ***3***
10.	
11.	*6, 5, 4,* ***4****, 3*
12.	*6, 5,* ***5****, 4, 3*
13.	
14.	*6,* ***6****, 5, 4, 3*
15.	***7****, 6, 5, 4, 3*

The last one was tough. You decide to back off three sessions and start building up again after a day off:

16.	*6, 5, 4,* ***4****, 3*
17.	*6, 5,* ***5****, 4, 3*
18.	*6,* ***6****, 5, 4, 3*
19.	
20.	***7****, 6, 5, 4, 3*

You reached the sets and reps which challenged you last time. This time they are easier. Not easy, but not at your limit. Keep going.

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21.	*7, 6, 5, 4,* ***4***
22.	*7, 6, 5,* ***5****, 4*
23.	
24.	*7, 6,* ***6****, 5, 4*

You really had to push it to make the reps. Time to end the cycle. Rest for two to three days, then test yourself. You are likely to make it to eight. Converting a 6RM to an 8RM in less than a month is a significant improvement for an experienced strength athlete.

25.	
24.	***8RM***

If you are confident that you can increase your 1RM, take a different approach. Instead of the above, after two days of rest after the (7, 6, 6, 5, 4) session work up to a couple of easy singles and loaded cleans.

E.g., if 44kg was your old 1RM and you have been doing the program with 36kg, do this:

1. *Warm up in the manner you are accustomed to.*
2. *32kg x 3/2*
3. *36kg x 2/2*
4. *40kg x 1/2*
5. *Loaded clean, 44kg x 1*
6. *Loaded clean, 44kg x 1*

After two days of rest test your 1RM, as well as RM with 36kg. Repeat the last session, then attempt pressing 48kg after 5min of rest. After 5 more minutes do max reps with 36kg.

The last days of the cycle would then look like this:

24.	*7, 6,* ***6****, 5, 4*
25.	
24.	*Peaking/tapering session*
25.	
26.	***48kg x 1RM, 36kg x RM***

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Kettlebell Strength Plan 414B

See plan 414A for progressing, backing off, and testing details.

You will be using two weights:

1RM	Weight I 6-8RM	Weight II 4-5RM
48kg	36kg	40kg
44kg	36kg	40kg
40kg	32kg	36kg
36kg	28kg	32kg
24kg	20kg	22kg
22kg	18kg	20kg
20kg	16kg	18kg

Each phase is two weeks long. Regardless of where you are, on day fifteen—Monday two weeks later—switch to the heavier kettlebell. Use the same type of progression, although with lower reps, for another week and a half. Then take two days off and test on Saturday of week four.

Days	Training
1-14	Weight I
15-24	Weight II
25-26	Off
27	**Test**
28	Off

Kettlebell Strength Plan 414C

See plan 414A for progressing, backing off, and testing details.

As in plan 414B, train with two weights, 6-8RM and 4-5RM.

This plan differs from the last one in its manner of cycling: you are pressing the heavier weight on one day, the lighter one next day, and keep rotating.

Press 2-3 days in a row, then take a day off (occasionally two). Another option: weight I on Mondays and Thursdays and weight II on Tuesdays and Fridays.

When the going gets tough or you start feeling fatigue building up, back off to the reps specified 2-3 sessions ago. Do if for both weights, even if only one is stalling. E.g.:

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Days	***Weight I***	***Weight II***
1.	*6, 5, 4, 3, 2*	
2.		*3, 2, 1, 1*
3.	*6, 5, 4, 3, **3***	
4.	*Off*	
5.		*3, 2, **2**, 1*
6.	*6, 5, 4, **4**, 3*	
7.	*Off*	
8.		*3, **3**, 2, 1*
9.	*6, 5, **5**, 4, 3*	
10.	*Off*	
1.	*Off*	
12.		***4**, 3, 2, 1*
13.	*6, **6**, 5, 4, 3*	
14.		*4, 3, 2, 1, **1***
15.	*Off*	
16.	***7**, 6, 5, 4, 3*	
17.	*Off. The last session took a lot of effort. Take 3 steps back and carry on.*	
18.		*3, 2, **2**, 1*
19.	*6, 5, 4, **4**, 3*	
20.	*...and so on.*	

When it takes an RPE 9 effort to complete your reps in perfect form with either weight, take two days of rest, then taper:

1. Warm up in the manner you are accustomed to
2. Weight I x 2/3 (reps/sets)
3. Weight II x 1/2
4. Loaded clean with your old 1RM x 1/2
5. Loaded clean with a weight heavier than your old 1RM x 1/2

Take two more days of rest and test:

1. Warm up in the manner you are accustomed to.
2. Weight I x 2/2 (reps/sets)
3. Weight II x 1/2
4. Your old 1RM x 1
5. Attempt a new 1RM or do several reps with your old 1RM.

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Power to the People! 2.1

(Zonin, based on Tsatsouline)

In this example we will use four lifts:

1. Back squat
2. Bench press
3. Deadlift
4. Military press

Of course, you may plan for a different number of lifts.

Weekly schedule

You will train each lift 2-3 times a week.

The tables below show two examples of training templates, one with constant training frequency, the other with variable training frequency.

Example 1						
Mon	***Tue***	***Wed***	***Thu***	***Fri***	***Sat***	***Sun***
BP	*MP*		*BP*	*MP*		
SQ	*DL*		*SQ*	*DL*		

Example 2							
Week	***Mon***	***Tue***	***Wed***	***Thu***	***Fri***	***Sat***	***Sun***
1	*BP*	*MP*		*BP*	*MP*		
	SQ	*DL*		*SQ*	*DL*		
2	*BP*	*MP*	*BP*		*MP*	*BP*	
	SQ	*DL*	*SQ*		*DL*	*SQ*	
3	*MP*		*BP*		*MP*		
	DL		*SQ*		*DL*		

Identify your starting weight "X"

Load the bar with a weight that feels very light (RPE 3-4).

1. Perform 10 reps.
2. Rest no less than 3min.
3. Add some weight to the bar.
4. Repeat steps 1-3 until the barbell slows, even slightly, on the last rep (RPE 7-8).

The load of your last set is your "X" weight.

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Identify your weight jump "Y"

Weight jump "Y"	
X	**Y**
<50kg/100lb	2.5kg/5lb every 2 sessions
50-100kg/100-200lb	2.5kg/5lb every session
>100kg/200lb	5kg/10lb every session

Progression

- Start the cycle with the weight "X" that you have established for each lift.
- Perform 5 sets of 5 reps (5/5).
- Every 1-2 sessions increase the load in a linear fashion by always adding the same amount of weight, "Y."
- Continue to add weight relentlessly until you reach the point of not being able to complete any of the sets of 5 reps perfectly. Make sure you terminate a set before failing the rep!
- Skip the next session scheduled for the lift in which you have been unable to do a perfect 5/5.
- Start the cycle over again, with the initial weight heavier by "Y."
- Run the cycle independently for each lift, as it is very likely that they will progress at a different pace.
- When you stop having improvements in one lift from one cycle to the next, it's time to move on to a new program.

Example:

- *Lift: bench press*
- *X = 60kg*
- *Y = 2.5kg*
- *Weekly template: example 2*

Week	*Mon*	*Wed*	*Thu*	*Fri*
1	*5/5 @ 60kg*		*5/5 @ 62.5kg*	
2	*5/5 @ 65kg*	*5/5 @ 67.5kg*		*5/5 @ 70kg*
3		***5,5,4 @ 72.5kg****		
4	*Rest*		*5/5 @ 62.5kg*	
5	*5/5 @ 65kg*	*5/5 @ 67.5kg*		*5/5 @ 70kg*
6		***5/5 @ 72.5kg*****		
7	*5/5 @ 75kg*		*5/5 @ 77.5kg*	
8	***5,3 @ 72.5kg****	*Rest*		*5/5 @65kg*
…				

**Failed to complete 5 perfect reps.*
***Completed 5 reps with the weight with which the athlete has previously failed.*

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Barbell Plan ROP-H

(Zonin)

In this cycle the main training weight remains the same for 4 weeks and then increases, hence at first glance it might appear to be a step cycle.

However, due to the way the progression is structured, the ARI increases each week, which qualifies this plan as a wave cycle.

SESSION A *(e.g., Monday)*		**SESSION B** *(e.g., Tuesday)*		**SESSION C** *(e.g., Thursday)*		**SESSION D** *(e.g., Friday)*	
Vertical press	H	Hip hinge	H	Horizontal press	H	Squat	H
Vertical pull				Horizontal pull			
Horizontal press	L	Squat	L	Vertical press	L	Hip hinge	L
Horizontal pull				Vertical pull			
Vertical press	SV	Hip hinge	SV	Horizontal press	SV	Squat	SV
Vertical pull				Horizontal pull			

If you have the luxury of occupying two barbells at the same time you may pair up the vertical press with the vertical pull and the horizontal press with the horizontal pull exercises and alternate one set of the first with one set of the second. If you do so, make sure you rest at least 3min between sets of the same exercise.

Step 1: Assign the main and the SV exercises to each movement pattern

Example:

SESSION A		***SESSION B***		***SESSION C***		***SESSION D***	
Military press	*H*	*Deadlift*	*H*	*Bench press*	*H*	*Back squat*	*H*
Pullup				*Pendlay row*			
Bench press	*L*	*Back squat*	*L*	*Military press*	*L*	*Deadlift*	*L*
Pendlay row				*Pullup*			
Double KB MP	*SV*	*RDL*	*SV*	*Dip*	*SV*	*Double KB FSQ*	*SV*
Chin-up				*Gorilla row*			

Step 2: Establish the weekly jump "Y"

Follow the Reload 2.0 table and instructions.

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Step 3: Outline the heavy day progression

1. Fill in the table according to the instructions indicated in each cell.
2. Round the weights you calculated to the nearest weight you have available.

Heavy day			
Week	**Heavy weight**	**Medium weight**	**Reps/Sets**
1	**1RM – (Yx5)**	1RM – (Yx7)	(2, 3, 5)/3
2			(**2**, 3, 5)/3
3			(**2**, **3**, 5)/3
4			(**2**, **3**, **5**)/3
5	**1RM – (Yx4)**	1RM – (Yx6)	(2, 3, 5)/3
6			(**2**, 3, 5)/3
7			(**2**, **3**, 5)/3
8			(**2**, **3**, **5**)/3
9	**1RM – (Yx3)**	1RM – (Yx5)	(2, 3, 5)/3
10			(**2**, 3, 5)/3
11			(**2**, **3**, 5)/3
12			(**2**, **3**)/3
13	**1RM – (Yx2)**	1RM – (Yx4)	(**1**, 3)/2
14			**1**/3
15	**1RM TEST**		
The sets highlighted in **bold** are done with the heavy weight, the rest with the medium weight.			

Example:

- *Bench press*
- *1RM = 110kg*
- *Weekly jump Y = 4.8kg*
- *Smallest weight jump available = 2.5kg*

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Heavy day			
Week	**Heavy weight**	**Medium weight**	**Reps/Sets**
1	**85kg**	75kg	(2, 3, 5)/3
2			(**2**, 3, 5)/3
3			(**2**, **3**, 5)/3
4			(**2**, **3**, **5**)/3
5	**90kg**	80kg	(2, 3, 5)/3
6			(**2**, 3, 5)/3
7			(**2**, **3**, 5)/3
8			(**2**, **3**, **5**)/3
9	**95kg**	85kg	(2, 3, 5)/3
10			(**2**, 3, 5)/3
11			(**2**, **3**, 5)/3
12			(**2**, **3**)/3
13	**100kg**	90kg	(**1**, 3)/2
14			**1**/3
15	**1RM TEST**		

*The sets highlighted in **bold** are done with the heavy weight, the rest with the medium weight.*

*E.g., on week 7 the table prescribes (**2**, **3**, 5)/3 with **90kg** as the heavy weight and 80kg as the medium weight. That means repeat 90kg x 2, 90kg x 3, 80kg x 5 three times.*

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Step 4: Outline the light day progression

1. Fill in the table according to the instructions indicated in each cell.
2. Round the weights you calculated to the nearest weight available.

Light day		
Week	**Weight**	**Reps/Sets**
1	60% 1RM	5/5
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13	60% 1RM + Y	4/5
14	60% 1RM + (Yx2)	3/5
15	60% 1RM + (Yx3)	2/5

Same example:

- *Bench press*
- *1RM = 110kg*
- *Smallest weight jump available = 2.5kg*

Light day		
Week	***Weight***	***Reps/Sets***
1	*65kg*	*5/5*
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13	*70kg*	*4/5*
14	*75kg*	*3/5*
15	*80kg*	*2/5*

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Specialized variety

I'm leaving up to you the management of the above.

Minimalist versions

It is possible to design minimalist versions of the Barbell ROP-H based on two or three lifts and no specialized variety exercises. Sime examples of minimalist schedules are below.

SESSION A *(e.g., Monday)*		**SESSION B** *(e.g., Wednesday)*		**SESSION C** *(e.g., Friday)*	
Upper body press	H	Squat / Hip hinge	H	Upper body pull	H
Upper body pull	L	Upper body press	L	Squat / Hip hinge	L

SESSION A *(e.g., Monday)*		**SESSION B** *(e.g., Thursday)*	
Upper body press / pull	H	Squat / Hip hinge	H
Squat / Hip hinge	L	Upper body press / pull	L

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Reload

Step 1: Test your 1RM

1. Test your 1RM in all the lifts you wish to plan, ensuring that the technique is perfect.

Step 2: Test your RM@80% 1RM and calculate your weekly jump and ramp-up baseline

Perform this test for each one of the lifts you wish to plan, a couple of days after you have tested your 1RM.

1. Calculate 80% 1RM of each one of the lifts you wish to plan and round it to the closest weight available.
2. Perform as many perfect reps as possible with 80% 1RM (rounded) of each one of the lifts.
3. Use the following table to determine your weekly jump, expressed in % 1RM, according to the number of perfect reps you have completed with 80% 1RM.
4. Calculate your weekly jump, expressed in kilograms or pounds, and round it to the closest weight available in your training facility.
5. Use the following table to determine your starting weight for a Reload cycle, expressed in % 1RM, according to the number of perfect reps you have completed with 80% 1RM.

RM@80% 1RM	Weight jump, % 1RM	Week 1 weight, % 1RM
≤5	5%	60%
6-8	4%	65%
9-10	3%	70%
>10	2%	75%

6. Calculate your cycle starting weight, expressed in kilograms or pounds, and round it to the closest weight available in your training facility.

Step 3: Find your 5/5@#5

Perform this test for each one of the lifts you wish to plan, a couple of days after you have tested your RM @ 80%1RM. You must start the test on each lift with the cycle starting weight you calculated in step two, and increase the weight, from set to set, by an amount that equals the weekly jump you also calculated in step two.

1. Perform 5 reps with the cycle starting weight.
2. Rest at least 3 minutes.
3. Add a weight that equals your weekly jump.
4. Perform 5 reps.
5. Repeat steps 2 through 4 until you cannot do a set of 5 perfect reps. This means if you have completed 5 reps, but even one has imperfect form, then you should consider the set unsuccessful.

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The heaviest weight with which you complete 5 perfect reps is your 5/5@#5 goal—the target weight for 5/5 in week 5.

Step 4: Lay down your 8-week plan

Once you have determined your weekly jump and set your 5/5@#5 goal, you have all the data you need to design a custom plan. You will need to repeat the following process for each one of the lifts you wish to plan.

1. Assign 5 sets of 5 reps to each week from week 1 to week 5.
2. Assign 3 sets of 3 reps to week 6.
3. Assign 2 sets of 2 reps to week 7.
4. Set week 8 as the new 1RM test week.
5. Assign your 5/5@#5 goal as the training load of week 5.
6. Count backward from week 5 to week 1 and subtract, week by week, your weekly jump from the training load.
7. Count forward from week 5 to week 7 and add, week by week, your weekly jump to the training load.

And there you go; you have your customized Reload cycle!

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Reload 2.0

(Zonin)

The updated version of Reload differs from the original as follows:

1. The weight jump algorithm is more precise; the table reflects it.
2. The training load is rounded week by week instead of once.
3. Instead of a 5/5 goal on week 5, the weekly load is calculated by counting back from week 8.
4. The training schedule changes in the peaking phase.
5. OTM singles are introduced in weeks 6-8.

Reload 2.0 algorithm

When designing cycles for beginners and intermediates there is no need for extreme precision when establishing the weight jump in the progression. However, the necessary level of precision increases exponentially as the athlete progresses to advanced. Many athletes, even elite ones, have missed a peak performance because their weekly jumps were either too small or too large.

Over the past 3 years at StrongFirst we have developed and tested a new algorithm that allows for a more accurate calculation of weekly load increments based on RM@80% 1RM. The table below compares Reload 1.0 and Reload 2.0.

RM@80% 1RM	Jump, % 1RM	
	Reload 1.0	Reload 2.0
4	5%	8.9%
5		6.7%
6	4%	5.3%
7		4.4%
8		3.8%
9	3%	3.3%
10		3.0%
11	2%	2.7%
12		2.4%

The new and refined table allows us to assign the weekly loads more precisely.

Let us use a practical example to compare the old and the new procedure and the Reload 1.0 and 2.0. Let's assume we are dealing with an athlete with the following numbers in the deadlift:

- *1RM = 225kg*
- *80% 1RM = 180kg*
- *RM@80% 1RM = 8*
- *Target weight on week 5 = 185kg*

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Here is what we would do with the 1.0 procedure. According to the table, the weekly jump is 4% and the equivalent value in kg is 9kg (225kg x 4% = 9). Let's say the smallest plates available in the training facility are 1.25kg, hence the minimum weight jump is 2.5kg. The weekly jump rounded value is 10kg.

With the 2.0 procedure, instead of rounding the weekly jump calculated according to the table to the closest weight available, we use it as is to fill in the table. Then, we round every single weekly load to the closest weight available in the training facility.

According to the Reload 2.0. table, the weekly jump is 3.8% and the equivalent value in kg is 8.6kg (225kg x 3.8% = 8.6).

The table below compares the original and the new procedures.

Week	**Reps/Sets**	**Reload 1.0**	**Reload 2.0**	
		Weight	**Weight**	**Weight** *(rounded)*
1	5/5	145	149	150
2	5/5	155	158	157.5
3	5/5	165	167	167.5
4	5/5	175	176	175
5	**5/5**	**185**	**185**	**185**
6	3/3	195	194	195
7	2/2	205	203	202.5
8	**1RM test**			

The following table will make the calculations easier. "Y" is the weekly weight jump.

Week	**Weight**
1	1RM – (8 x Y)
2	1RM – (7 x Y)
3	1RM – (6 x Y)
4	1RM – (5 x Y)
5	1RM – (4 x Y)
6	1RM – (3 x Y)
7	1RM – (2 x Y)

E.g., a lifter's bench press numbers:

1. *1RM = 110kg*
2. *Weekly jump = 4.8kg*
3. *Smallest weight jump available = 2.5kg*

On week 1, he is supposed to lift (1RM – 8 x weekly jump).

8x4.8=38.4
110-38.4=71.6
Round up to use 2.5kg plates: 72.5kg

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Do the math for the entire cycle:

Heavy day		
Week	***Weight***	***Reps/Sets***
1	*72.5kg*	*5/5*
2	*77.5kg*	*5/5*
3	*80kg*	*5/5*
4	*85kg*	*5/5*
5	*90kg*	*5/5*
6	*95kg*	*3/3*
7	*100kg*	*2/2*
8	***1RM TEST***	

Reload 2.0 schedule

As in the original Reload, in weeks 1-5 you build a foundation of strength for peaking in weeks 6-8.

Train a lift twice a week, a heavy day (H) and a light day (L) on weeks 1-5.

The heavy day's sets and reps and progression are identical to those in Reload 1.0, except the weights are assigned according to the new algorithm.

The light day stays the same throughout the cycle: 60% 1RM x 5/5.

In weeks 6-8, a third training day is added, a singles day (S). You are to do 5-10 singles with 85% 1RM on the minute. They will hone your single rep skill, especially the set-up, for the upcoming 1RM test.

Because after a maximal or near-maximal muscular recruitment of the nervous system stays in a highly excited state for 2min, you will be surprised at how easily you will be able to complete all the reps, and how the ones after the first one will feel lighter.

<table>
<tr><th colspan="5">Reload 2.0 cycle</th></tr>
<tr><th colspan="3">Heavy day</th><th>Light day</th><th>Singles day</th></tr>
<tr><th>Week</th><th>Weight</th><th>Reps/Sets</th><th colspan="2">Weight x Reps/Sets</th></tr>
<tr><td>1</td><td>1RM – (8 x Y)</td><td rowspan="5">5/5</td><td rowspan="8">60% 1RM x 5/5</td><td rowspan="5">N/A</td></tr>
<tr><td>2</td><td>1RM – (7 x Y)</td></tr>
<tr><td>3</td><td>1RM – (6 x Y)</td></tr>
<tr><td>4</td><td>1RM – (5 x Y)</td></tr>
<tr><td>5</td><td>1RM – (4 x Y)</td></tr>
<tr><td>6</td><td>1RM – (3 x Y)</td><td>3/3</td><td>85% 1RM x 1/10</td></tr>
<tr><td>7</td><td>1RM – (2 x Y)</td><td>2/2</td><td>85% 1RM x 1/7</td></tr>
<tr><td>8</td><td colspan="2">1RM TEST</td><td>85% 1RM x 1/5</td></tr>
</table>

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Training multiple lifts

Now that you understand how the cycle works, you can design one for multiple lifts. Feel free to include specialized variety exercises. Following is a sample schedule.

WEEKS 1-5							
SESSION A *(e.g., Monday)*		**SESSION B** *(e.g., Tuesday)*		**SESSION C** *(e.g., Thursday)*		**SESSION D** *(e.g., Friday)*	
Bench press	H	Deadlift	H	Military press	H	Back squat	H
Back squat	L	Bench press	L	Deadlift	L	Military press	L

WEEKS 6-8					
SESSION A *(e.g., Monday)*		**SESSION B** *(e.g., Wednesday)*		**SESSION C** *(e.g., Friday)*	
Back squat	S	Back squat	L	Back squat	H
Bench press		Bench press		Bench press	
Deadlift		Deadlift		Deadlift	
Military press		Military press		Military press	

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The Battleship (BtS-RND)

(Zonin)

This is a Built Strong plan based on variable overload in which the weekly load is determined by two rolls of a die.

Step 1: Test your 1RMs

Test your 1RM in each of the lifts you wish to train.

Step 2: Establish your training weights

Calculate the following % 1RM for each lift: 85, 75, 65. Round the results to the closest weight available in your training facility.

Step 3: Test your RMs with each training weight

Test how many perfect reps you can complete with each weight you have calculated.

Step 4: Establish your rep ladders

Use the table below to establish your rep ladders according to the RM with each weight.

RM	Ladder
4	1, 2, 3
5	2, 3, 3
6	2, 3, 4
7	2, 4, 5
8	3, 4, 5
9	3, 5, 6
10	3, 5, 7
11	4, 6, 7
12	4, 6, 8
13	4, 7, 9
14	5, 7, 9
15	5, 8, 10

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Example:

- *Bench press*
- *1RM = 110kg*

- *% 1RM:*
  - *85% 1RM = 93.5kg*
  - *75% 1RM = 82.5kg*
  - *65%1RM = 71.5kg*

- *% 1RM rounded:*
  - *85% 1RM = 92.5kg*
  - *75% 1RM = 82.5kg*
  - *65% 1RM = 72.5kg*

- *RMs:*
  - *5RM @ 92.5kg*
  - *9RM @ 82.5kg*
  - *13RM @ 72.5kg*

- *Ladders:*
  - *92.5kg: 2, 3, 3*
  - *82.5kg: 3, 5, 6*
  - *72.5kg: 4, 7, 9*

Weekly templates

Below are some examples of possible weekly schedules, for 3, 4, and 6 lifts.

3 lifts, 3 sessions/week

SESSION A		SESSION B		SESSION C	
Upper body press	H	Squat	H	Upper body pull	H
Squat	M	Upper body pull	M	Upper body press	M
Upper body pull	L	Upper body press	L	Squat	L

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4 lifts, 3 sessions/week

SESSION A		SESSION B		SESSION C	
Upper body press	H	Upper body pull	H	–	H
Hip hinge		Squat			
Upper body pull	M	–	M	Upper body press	M
Squat				Hip hinge	
–	L	Upper body press	L	Upper body pull	L
		Hip hinge		Squat	

If you have the luxury of occupying two barbells at the same time you may pair up the upper body press with the hip hinge and the upper body pull with the squat and alternate one set of the first with one set of the second. If you do so, make sure you rest at least 3min between sets of the same exercise.

4 lifts, 4 sessions/week

SESSION A		SESSION B		SESSION C		SESSION D	
Upper body press	H	Squat	H	Upper body pull	H	Hip hinge	H
Upper body pull	M	Hip hinge	M	Upper body press	M	Squat	M
Hip hinge	L	Upper body press	L	Squat	L	Upper body pull	L

6 lifts, 4 sessions/week

SESSION A		SESSION B		SESSION C		SESSION D	
Horizontal press	H	Squat	H	Vertical press	H	Hip hinge	H
Horizontal pull				Vertical pull			
Vertical press	M	Hip hinge	M	Horizontal press	M	Squat	M
Vertical pull				Horizontal pull			
Hip hinge	L	Horizontal press	L	Squat	L	Vertical press	L
		Horizontal pull				Vertical pull	

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If you have the luxury of occupying two barbells at the same time you may pair up the vertical press with the vertical pull and the horizontal press with the horizontal pull exercises and alternate one set of the first with one set of the second. If you do so, make sure you rest at least 3min between sets of the same exercise.

How to execute the plan

At the beginning of each week, roll a die twice for each lift (if you don't have a die, download a die-roll app on your phone).

The weekly volume of each lift in each intensity zone will depend on the result of the die rolls and is outlined in the table below.

		FIRST ROLL							
		1		2-3		4-5		6	
SECOND ROLL		DAY	NL	DAY	NL	DAY	NL	DAY	NL
	1	H	6	H	6	H	6	H	6
		M	21	M	34	M	44	M	57
		L	33	L	48	L	62	L	77
		ARI	70,5%	ARI	70,2%	ARI	70,0%	ARI	69,9%
	2-3	DAY	NL	DAY	NL	DAY	NL	DAY	NL
		H	9	H	9	H	9	H	9
		M	18	M	31	M	41	M	54
		L	33	L	48	L	62	L	77
		ARI	71,0%	ARI	70,6%	ARI	70,3%	ARI	70,1%
	4-5	DAY	NL	DAY	NL	DAY	NL	DAY	NL
		H	11	H	11	H	11	H	11
		M	16	M	29	M	39	M	52
		L	33	L	48	L	62	L	77
		ARI	71,3%	ARI	70,8%	ARI	70,4%	ARI	70,3%
	6	DAY	NL	DAY	NL	DAY	NL	DAY	NL
		H	14	H	14	H	14	H	14
		M	13	M	26	M	36	M	49
		L	33	L	48	L	62	L	77
		ARI	71,8%	ARI	71,1%	ARI	70,7%	ARI	70,5%

**If the result of your die rolls leads to the same cell as the previous week, roll again!*

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Same example: bench press

You are training 3 lifts and doing 3 sessions/week hence you are training your bench press in the following manner:

- *Session 1: Heavy (85% 1RM, 92.5kg)*
- *Session 2: Light (75% 1RM, 82.5kg)*
- *Session 3: Medium (65% 1RM, 72.5kg)*

You rolled a 2 and a 6. Your weekly volumes are:

- *Session 1: Heavy, NL 14*
- *Session 2: Light, NL 26*
- *Session 3: Medium, NL 48*

Your breakdown in sets and reps:

- *Session 1: Heavy, 2, 3, 3, 2, 3, 1 (NL 14)*
- *Session 2: Light, 3, 5, 6, 3, 5, 4 (NL 26)*
- *Session 3: Medium, 4, 7, 9, 4, 7, 9, 4, 4 (NL 48)*

Behind the scenes

The first roll establishes the total volume of the week. The difference among the rolled volumes ensures compliance with the Delta 20 principle.

Die shows	**1**	**2-3**	**4-5**	**6**	**Total**
Weekly NL	60	88	112	140	400

55% of the weekly volume is represented by the light sessions.

Weekly NL	**60**	**88**	**112**	**140**	**Total**
Light sessions' NL (55%)	33	48	62	77	220

The second roll establishes the % of the total NL assigned to the heavy sessions. This allows to wave the ARI from 69.9% to 71.8% 1RM.

Die shows	**1**	**2-3**	**4-5**	**6**
% NL assigned to the heavy session	1.5%	2.2%	2.8%	3.5%
Light sessions' NL (rounded)	6	9	11	14

The volume of the medium sessions is obtained by subtracting the heavy and light session's NLs from the total weekly NL.

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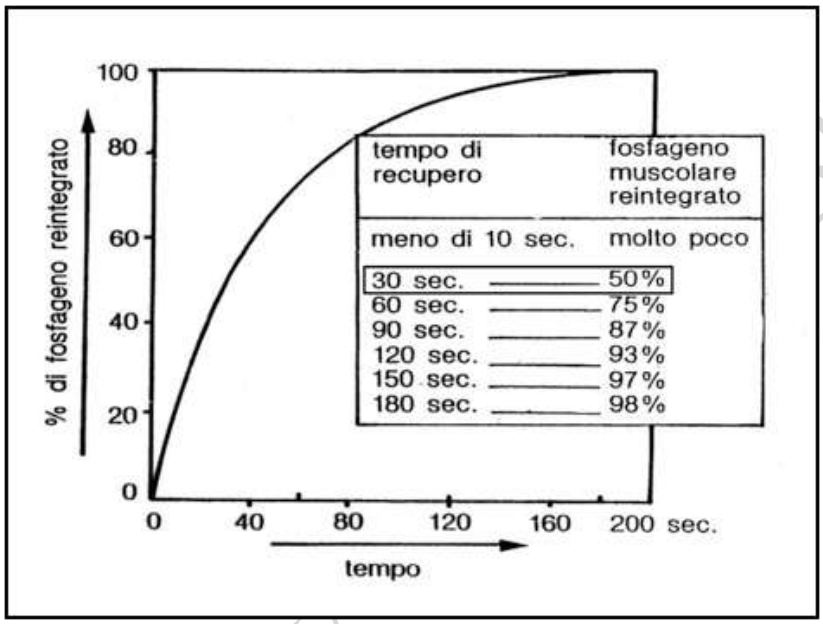
Acute Supercompensation and the HYP 12 Method

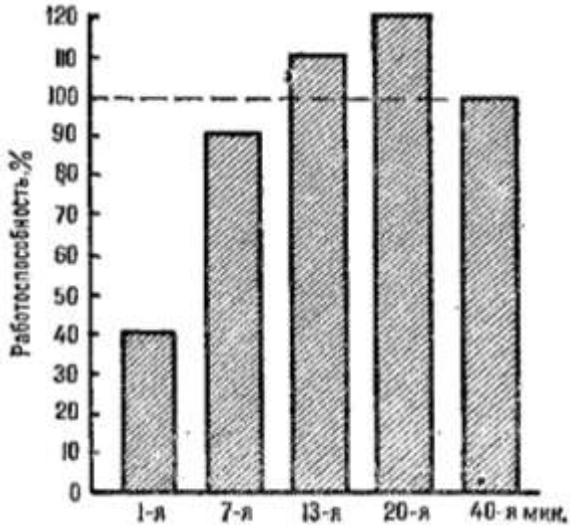
(Zonin)

It is it is almost universally acknowledged that full recovery of phosphates (ATP and CP) requires about 3min. However, at 3min the recovery has not yet reached 100% but around 98%. It takes about 10min for complete recovery.



It is interesting to note that after only 30 seconds of recovery, the phosphate reserves have already been replenished by around 50%. This provides a rationale for techniques such as rest-pause, cluster training, etc., which that are very commonly used in bodybuilding.

But what happens if recovery lasts longer than 10min? Let us consider the chart below.



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Soviet athletes performed a set of military presses to refusal. Their RMs was recorded as 100% on the Y axis. For simplicity let us say they have completed 10 repetitions (10RM). They then performed a second set to TMI after having recovered for 1min.

They repeated the same test consisting of 2 sets conducted to TMI on different days, each time varying the recovery time between the first and second set.

Recovery times were recorded on the X-axis of the graph. Note that the vertical bars in the graph refer to the second sets, tested on different days.

As shown in the graph, after 1min of rest the subjects were barely able to complete 4 reps. As expected, after 10min the subjects were able to repeat the performance of the first set, completing 10 reps. Then things got really interesting.

After 20min the athletes exceeded their RM of the first set and completed 12 repetitions. An increase of 20%! This supercompensation provides the rationale behind the HYP 12 method. A practical application of this experiment is outlined in the following HYP 12 Wave Cycle.[7]

This 15-week wave strength and hypertrophy cycle implements a strategy that takes advantage of the acute supercompensation of phosphates.

You will train the main lifts twice a week, with one heavy day (H) plus sets to refusal (R) and one light day (L). You may also add upper body pulls, midsection work, etc.

SESSION A *(e.g., Monday)*		**SESSION B** *(e.g., Tuesday)*		**SESSION C** *(e.g., Thursday)*		**SESSION D** *(e.g., Friday)*	
Bench press	H	Deadlift	H	Military press	H	Back squat	H
Back squat	L	Bench press	L	Deadlift	L	Military press	L
Bench press	R	Deadlift	R	Military press	R	Back squat	R
H = heavy day; L = light day; R = sets to refusal day							

Below is how to design your HYP 12 Wave Cycle.

Note that you will have to repeat the procedure outlined below for each main lift.

Step 1: Establish the weekly weight jump "Y"

Follow the Reload 2.0 table and instructions.

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[7] Another is featured in the "Six Weeks to Strong: The SF 930 Plan" article on StrongFirst.com.

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Step 2: Outline the heavy day progression

1. Fill in the table according to the instructions indicated in each cell.
2. Round the weights you have calculated to the nearest weight you have available.

Heavy Day		
Week	**Weight**	**Reps/Sets**
1	1RM – (Yx8)	5/5
2	1RM – (Yx7)	4/5
3	1RM – (Yx6)	3/5
4	1RM – (Yx7)	5/5
5	1RM – (Yx6)	4/5
6	1RM – (Yx5)	3/5
7	1RM – (Yx6)	5/5
8	1RM – (Yx5)	4/5
9	1RM – (Yx4)	3/5
10	1RM – (Yx5)	5/5
11	1RM – (Yx4)	4/5
12	1RM – (Yx3)	3/5
13	1RM – (Yx2)	(1, 2)/2
14	1RM – Y	1/4
15	**1RM TEST!**	

Example:

- *Bench press*
- *1RM = 110kg*
- *Weekly jump Y = 4.8kg*
- *Smallest weight jump available = 2.5kg*

Heavy Day		
Week	***Weight***	***Reps/Sets***
1	*72.5kg*	*5/5*
2	*77.5kg*	*4/5*
3	*80kg*	*3/5*
4	*77.5kg*	*5/5*
5	*80kg*	*4/5*
6	*85kg*	*3/5*
7	*80kg*	*5/5*
8	*85kg*	*4/5*
9	*90kg*	*3/5*
10	*85kg*	*5/5*
11	*90kg*	*4/5*
12	*95kg*	*3/5*
13	*100kg*	*(1, 2)/2*
14	*105kg*	*1/4*
15	***1RM TEST!***	

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Step 3: Outline the light day progression

1. Fill in the table according to the instructions indicated in each cell.
2. Round the weights you have calculated to the nearest weight you have available.

Light Day		
Week	**Weight**	**Reps/Sets**
1	60% 1RM	5/5
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13	60% 1RM + Y	4/5
14	60% 1RM + (Yx2)	3/5
15	60% 1RM + (Yx3)	2/5

Same example:

- *Bench press*
- *1RM = 110kg*
- *Weekly jump Y = 4.8kg*
- *Smallest weight jump available = 2.5kg*

Light Day		
Week	***Weight***	***Reps/Sets***
1	*65kg*	*5/5*
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13	*70kg*	*4/5*
14	*75kg*	*3/5*
15	*80kg*	*2/5*

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Step 4: Outline the sets to refusal progression

1. Fill in the table according to the instructions indicated in each cell.
2. Round the weights you have calculated to the nearest weight you have available.

Sets to Refusal		
Week	**Weight**	**Reps/Sets**
1	1RM – (Yx8)	RM/3
2		
3		
4	1RM – (Yx7)	
5		
6		
7	1RM – (Yx6)	
8		
9		
10	1RM – (Yx5)	
11		
12		
13	—	—
14		
15		

Same example:

- *Bench press*
- *1RM = 110kg*
- *Weekly jump Y = 4.8kg*
- *Smallest weight jump available = 2.5kg*

Sets to Refusal		
Week	***Weight***	***Reps/Sets***
1	*72.5kg*	*RM/3*
2		
3		
4	*77.5kg*	
5		
6		
7	*80kg*	
8		
9		
10	*85kg*	
11		
12		
13	—	—
14		
15		

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All RS sets must be conducted to RM, or rather to *"refusal,"* meaning your muscles are so tired that they literally refuse to undertake another repetition. This of course does not authorize you to perform reps that are less than perfect.

How to execute the plan

Start each session with the heavy day of the prescribed lift (e.g., Session A: bench press).

Once you have completed the last heavy set, take at least 3min of rest, and then perform the first set to refusal for the same lift (e.g., Session A: bench press).

As soon as you have completed the set to refusal, immediately start a 12min countdown on your timer.

Continue your session with the light day of the prescribed lift (e.g., Session A: back squat).

As soon as the 12min countdown is over, and the timer rings, do the second set to refusal. If when the timer rings you are in the process of performing a set of the light lift, complete the set in progress, take a couple of minutes of rest, and then do the set to refusal. 12min is the minimum rest between the sets to refusal and it may be longer.

Start the 12min countdown again and resume your training from where you left it.

Once the 12min countdown is over, perform the third and final set to refusal.

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For further study

State-of-the-art online programming seminars:

- **Plan Strong™**—the undefeated Soviet Olympic weightlifting system applied to general strength training (the powerlifts, the barbell and kettlebell military presses, weighted calisthenics)

- **Built Strong**—the same system adapted for building muscle that is as strong as it looks

Power to you!